

Learning Reasoning Rewards from Expert Demonstrations with Inverse Reinforcement Learning

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Abstract

Teaching large language models (LLMs) to reason during post-training typically relies on reinforcement learning with explicit outcome- or process-based reward functions. However, in many real-world settings, obtaining or defining such reward functions is difficult, especially for complex tasks, making learning from expert demonstrations an attractive alternative. The dominant approach, supervised fine-tuning (SFT), trains models to imitate expert reasoning traces directly, but suffers from the general limitations of off-policy learning: performance can be fragile to inference-time deviations from states explicitly covered by the demonstrations. To address this, we propose **Reasoning Adversarial Inverse Reinforcement Learning (R-AIRL)**. Rather than imitating the expert’s reasoning, R-AIRL infers the underlying process-level reward from the expert Chain-of-Thoughts. Through experiments on GSM8K, MMLU-Pro and MedReason we show that the reasoning reward function learned with R-AIRL can be effectively used throughout the training and inference pipeline: (1) to provide a training signal for **post-training**, outperforming SFT in most of the considered settings, (2) for **inference-time reranking**, improving pass@1 by up to 17.4 points, and (3) for **process-level evaluation**, localising reasoning failures with up to 86.1% accuracy. Overall, R-AIRL bridges imitation learning and reward-based optimisation, enabling the extraction of meaningful reasoning signals from expert thinking traces.¹

1 Introduction

Reasoning post-training for large language models (LLMs) often relies on reinforcement learning (RL) with external outcome verifiers or step-level supervision [Guo et al., 2025, Uesato et al., 2022, Lightman et al., 2023]. In many practical settings, these signals are difficult to design or expensive to collect, whereas expert demonstrations (from humans or powerful LLMs) are comparatively easy to obtain. This makes supervised fine-tuning (SFT) on expert traces a particularly attractive approach in the demonstration-only regime [Guo et al., 2025]. However, SFT mainly trains models to simply imitate observed reasoning traces rather than to evaluate them or directly extrapolate from them. As a result, when generation deviates from demonstrated paths, the model lacks an explicit objective for comparing alternative reasoning steps or recovering from errors [Setlur et al., 2025].

A natural alternative is to optimise reasoning via a *learned* reward. Yet, manually specifying such a reward remains challenging: hand-crafted signals can be brittle, task-specific and difficult to scale. At the same time, expert demonstrations implicitly encode which trajectories are preferred [Abbeel and Ng, 2004, Ziebart et al., 2004]. This suggests learning a reward for reasoning directly from demonstrations, rather than prescribing one externally.

¹We provide the code for our experiments at https://github.com/fanconic/expert_reasoning

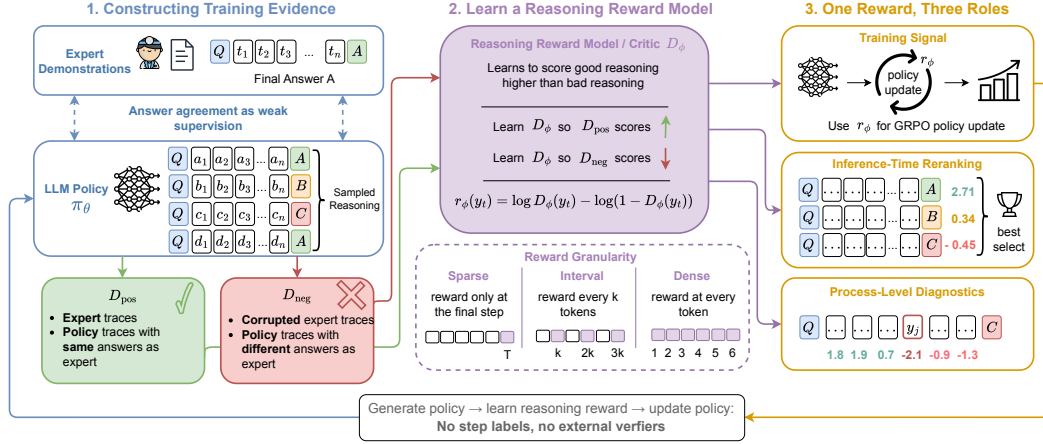


Figure 1: **R-AIRL: Learning and using reasoning rewards from expert demonstrations.** (i) We construct training evidence from expert and policy traces, (ii) learn a critic-based reward at sparse/interval/dense granularities, and (iii) reuse that reward for GRPO training, inference-time reranking, and process-level error localisation, without step labels or external verifiers.

This raises a key question: can we obtain a reward over reasoning traces that is both learnable from demonstrations and useful in practice? We argue that such a reward, in order to be truly useful, must satisfy three core requirements: it must provide a stable optimisation signal for policy learning, remain useful at inference time for selecting among candidate reasoning paths, and offer process-level feedback for diagnosing where reasoning fails.



We formalise three desiderata for a learned reasoning reward:

1. **(D1) Usable training signal.** A stable optimisation signal for on policy-generated trajectories.
2. **(D2) Inference-time utility.** Effective tool for ranking the sampled traces under a fixed budget, without retraining.
3. **(D3) Process-level feedback.** Step-level scoring function allowing to localise where reasoning first fails.

To satisfy these desiderata, we formulate reasoning as an inverse reinforcement learning (IRL) problem [Abbeel and Ng, 2004, Ziebart et al., 2004]. Our proposed method, Reasoning Adversarial Inverse Reinforcement Learning (R-AIRL), adapts adversarial IRL [Ho and Ermon, 2016, Fu et al., 2018] to the LLM reasoning setting: instead of teaching the model to imitate expert tokens, it learns a reasoning reward from demonstrations and then uses it to optimise the policy on its own sampled trajectories. In addition to providing a competitive training signal, the same reward can also be reused for inference-time reranking (to further boost performance) and fine-grained token/step diagnostics.

Contributions. We make three contributions:

1. **Learning rewards for reasoning.** We adapt adversarial IRL to learn rewards over reasoning trajectories from expert demonstrations at multiple levels of granularity.
2. **One reward, multiple roles.** We show that a single learned reward can be reused for policy optimisation, inference-time reranking, and error localisation.
3. **Empirical validation across training, inference, and transfer.** Across GSM8K, MEDREASON, and MMLU-PRO, learned rewards often outperform SFT, improve Best-of-16 reranking by up to 17.4 points, transfer across tasks/backbones in most settings, and localise reasoning errors.

Sections 5.1, 5.2, and 5.3 evaluate (D1)–(D3), respectively; Figure 1 summarises the full pipeline.

2 Problem Formalism

For a prompt $x \sim \mathcal{Q}$, we model an output as a reasoning trajectory $y = (y_1, \dots, y_T)$ with autoregressive likelihood $p_\theta(y \mid x) = \prod_{t=1}^T \pi_\theta(y_t \mid x, y_{<t})$, where $y_{<t}$ is the partial reasoning history.

Latent process reward. The true reward for reasoning is unobserved, so we model a process-level reward $r_\phi(y_t \mid x, y_{<t})$ and aggregate it as $R_\phi(y; m) = \frac{1}{T} \sum_{t=1}^T m_t r_\phi(y_t \mid x, y_{<t})$, with $m_t \in \{0, 1\}$. This unifies granularities: dense reward uses $m_t = 1$ for all t , while outcome/sparse reward is the special case $m_t = \mathbb{I}[t = T]$. The $1/T$ factor makes scores comparable across different trajectory lengths.

Inverse Reinforcement Learning (IRL). We treat expert demonstrations as samples from $p_E(y \mid x)$, i.e., from an unknown expert policy that is approximately optimal under a latent reasoning reward. We formulate reward learning as learning a process-level reward that separates expert from policy trajectories, and then update the policy to maximise that learned reward:

$$\max_{\phi} \min_{\theta} \mathbb{E}_{y^E \sim p_E} [R_\phi(y^E; m)] - \mathbb{E}_{y \sim \pi_\theta} [R_\phi(y; m)]. \quad (1)$$

The first term learns r_ϕ to prefer expert reasoning; the second drives policy optimisation under that reward. Because the expert term is independent of θ , minimising Eq. (1) w.r.t. θ is equivalent to maximising expected learned reward on policy trajectories (D1). Since r_ϕ is process-level, it yields token-level feedback (D3), and the same R_ϕ can be reused for inference-time reranking (D2).

3 Related Work

Using the notation from Section 2, we organise methods by supervision regime: verifier-based rewards, preference-based reward learning, and demonstration learning; we then compare whether they natively support (D1)–(D3).

Externally specified reward objectives. Outcome-supervised reasoning RL optimises the terminal-mask special case

$$\max_{\theta} \mathbb{E}_{y \sim \pi_\theta} [R(y; m^{\text{out}})], \quad m_t^{\text{out}} = \mathbb{I}[t = T]. \quad (2)$$

This reward is usually derived from an external verifier [Guo et al., 2025, Uesato et al., 2022, Yu et al., 2025, YuYue et al., 2025]. Process-reward models (PRMs) instead optimise $\max_{\theta} \mathbb{E}_{y \sim \pi_\theta} [R(y; m)]$ with denser masks m (e.g., full, interval, or step-wise). These approaches can be effective [Uesato et al., 2022, Lightman et al., 2023], and search-based variants can improve exploration [Zelikman et al., 2022, Yuan et al., 2023, Singh et al., 2024, Hosseini et al., 2024, Silver et al., 2017]. Relative to our setting, they typically provide strong optimisation signals (D1) and can support reranking (D2) when reward access is available, while (D3) depends on externally specified process supervision. Crucially, however, in R-AIRL we assume no access to an external verifier and instead learn the reasoning from demonstrations.

Preference-based reward learning. Preference-based RL learns a reward model R_ϕ , typically outcome-level, from pairwise preferences between a winning response y^w and a losing response y^l for the same prompt x . With $(x, y^w, y^l) \sim \mathcal{D}_{\text{pref}}$, a common objective is:

$$\max_{\phi} \mathbb{E} [\log \sigma(R_\phi(y^w; m^{\text{out}}) - R_\phi(y^l; m^{\text{out}}))], \quad \text{followed by} \quad \max_{\theta} \mathbb{E}_{y \sim \pi_\theta} [R_\phi(y; m^{\text{out}})].$$

This setting uses preference data rather than verifier-labelled rewards [Christiano et al., 2023, Rafailov et al., 2024]. This setting uses preference data rather than external verifier rewards [Christiano et al., 2023, Rafailov et al., 2024]. In our framing, preference-based rewards can support (D1) and (D2), but are typically outcome-level and therefore less direct for (D3). Again, a key distinction is that in R-AIRL, we assume no access to preference-labelled data and instead learn the reasoning from demonstrations.

Demonstration-only learning. When supervision is provided as expert trajectories, a common training paradigm is supervised fine-tuning (SFT), i.e., knowledge distillation on teacher/expert traces [Hinton et al., 2015], formalised through the following loss function:

$$\max_{\theta} \mathbb{E}_{y^E \sim p_E} \left[\frac{1}{T} \sum_t \log \pi_\theta(y_t^E) \right].$$

This often improves reasoning quality [Guo et al., 2025, Kang et al., 2023, Kujanpää et al., 2025, Xu et al., 2025], but it does not learn an explicit reward on policy-generated trajectories, so it is limited for ① and does not natively provide ② reranking or ③ token-level localisation [Setlur et al., 2025]. Recent self-distillation work in continual learning remains distillation-based and complementary to reward learning [Shenfeld et al., 2026]. Adversarial distillation variants [Ye et al., 2025, Lee et al., 2025] infer trajectory-level discriminator rewards from teacher traces as an additional learning signal together with an external verifier:

$$\max_{\phi} \min_{\theta} \mathbb{E}_{y^E \sim p_E} [R_{\phi}(y^E; m^{\text{out}})] - \mathbb{E}_{y \sim \pi_{\theta}} [R_{\phi}(y; m^{\text{out}})], \quad \max_{\theta} \mathbb{E}_{y \sim \pi_{\theta}} [R_{\phi}(y; m^{\text{out}}) + R(y; m^{\text{out}})].$$

R-AIRL differs by learning process-level scoring $R_{\phi}(y; m)$ from demonstrations and reusing one critic for ① training, ② reranking, and ③ token-level localisation.

We summarise this comparison in Table 1.

Method Class	Example Works	Optimisation Objective	Supervision Data	No External Verifier	D1	D2	D3
Verifier-based supervision							
Outcome-supervised RL	Guo et al. [2025]	$\max_{\theta} \mathbb{E}_{y \sim \pi_{\theta}} [R(y; m^{\text{out}})]$	Outcome verifier	✗	✓	✓	✗
	Yu et al. [2025]						
	YuYue et al. [2025]						
Process-supervised RL	Uesato et al. [2022]	$\max_{\theta} \mathbb{E}_{y \sim \pi_{\theta}} [R(y; m)]$	Process verifier	✗	✓	✓	✓
	Lightman et al. [2023]						
Preference supervision							
Preference-supervised RL	Christiano et al. [2023] Rafailov et al. [2024]	$\max_{\phi} \mathbb{E}_{(y^w, y^l) \sim \mathcal{D}_{\text{pref}}} \left[\log \sigma \left(R_{\phi}(y^w; m^{\text{out}}) - R_{\phi}(y^l; m^{\text{out}}) \right) \right]$ then $\max_{\theta} \mathbb{E}_{y \sim \pi_{\theta}} [R_{\phi}(y; m^{\text{out}})]$	Preference pairs	✗	✓	✓	✗
Demonstration supervision							
SFT	Hinton et al. [2015]	$\max_{\theta} \mathbb{E}_{y^E \sim \mathcal{D}_E} \left[\frac{1}{T} \sum_t \log \pi_{\theta}(y_t^E) \right]$	Demos	✓	✗	✗	✗
	Guo et al. [2025]						
	Xu et al. [2025]						
Adversarial distillation	Ye et al. [2025]	$\max_{\phi} \min_{\theta} (\mathbb{E}_{y^E \sim p_E} [R_{\phi}(y^E; m^{\text{out}})] - \mathbb{E}_{y \sim \pi_{\theta}} [R_{\phi}(y; m^{\text{out}})])$ then $\max_{\theta} \mathbb{E}_{y \sim \pi_{\theta}} [R_{\phi}(y; m^{\text{out}}) + R(y; m^{\text{out}})]$	Demos + outcome verifier	(✗)	✓	(✓)	✗
	Lee et al. [2025]						
R-AIRL	this work	$\max_{\phi} \min_{\theta} (\mathbb{E}_{y^E \sim p_E} [R_{\phi}(y^E; m)] - \mathbb{E}_{y \sim \pi_{\theta}} [R_{\phi}(y; m)])$	Demos	✓	✓	✓	✓

Table 1: **Related work comparison.** Methods are grouped by supervision regime (verifier-based, preference-based, demonstrations), with objectives, supervision sources, and verifier dependence. ①: reward-based training signal on policy traces. ②: inference-time reranking with the same reward. ③: token-level diagnostics for error localisation. Parenthesised checkmarks indicate partial support.

4 Method

Our goal is to learn a reasoning reward from expert demonstrations, so the policy can be optimised on its own rollouts without external verifiers. We build on the adversarial IRL view of GAIL/AIRL [Ho and Ermon, 2016, Fu et al., 2018], but with a different supervision construction: instead of source-based expert-vs-policy labels, we form verifier-free positive/negative trace sets from answer agreement with paired demonstrations and corrupted expert traces, train a token-level critic, and convert its logits into rewards for policy updates.

4.1 Training Evidence Construction

A naive split (all expert positive, all policy negative) encourages style discrimination rather than reasoning-quality discrimination [Ye et al., 2025, Lee et al., 2025]. We therefore use answer agreement with the paired expert trace as weak, verifier-free supervision. For each prompt, let y^E be the paired expert trace and let $\mathcal{O}(y)$ denote the answer tokens from generation y :

$$\mathcal{D}_{\text{pos}} = \{y \sim p_E\} \cup \{y \sim \pi_{\theta} \mid \mathcal{O}(y) = \mathcal{O}(y^E)\}.$$

Including answer-consistent policy traces in $\mathcal{D}_{\text{pos}}$ puts policy text on both sides, reducing source shortcuts and pushing the critic toward reasoning quality. For negatives, we use answer-mismatched policy traces and augment them with corrupted expert traces:

$$\mathcal{D}_{\text{neg}} = \{y \sim \pi_{\theta} \mid \mathcal{O}(y) \neq \mathcal{O}(y^E)\} \cup \{\mathcal{C}(y^E) \mid y^E \in \mathcal{D}^E\}.$$

Corruptions preserve expert style while injecting reasoning errors, further discouraging provenance-based separation. This supervision is intentionally weak: answer agreement can miss latent errors or penalise partially correct traces, but it provides enough signal to learn useful rewards from demonstrations alone, without step-level labels or pairwise preferences.

4.2 Reasoning Critic and Reward Definition

We train a token classifier $D_\phi(y_t) \in (0, 1)$ that assigns each token (and its previous context) a probability of belonging to the positive reasoning class ($\mathcal{D}_{\text{pos}}$ vs. $\mathcal{D}_{\text{neg}}$). Following Section 2, a binary mask $m_t \in \{0, 1\}$ specifies on which tokens supervision is applied along the reasoning trajectory. We evaluate three granularities: *Sparse Outcome* ($m_t = \mathbb{I}[t = T]$), *Interval* ($m_t = 1$ every k tokens), and *Fully Dense* ($m_t = 1$ for all t).

Granularity controls credit assignment: coarser masks are typically lower-variance and more stable for optimisation (D9), while denser masks provide more detailed process feedback for localisation (D3) but can be harder to optimise robustly. This enables a direct sparse-to-dense comparison using a single supervision source (demonstrations only), unlike pipelines that rely on external process labels or verifiers [Guo et al., 2025, Uesato et al., 2022, Lightman et al., 2023].

The critic objective is length-normalised binary cross-entropy:

$$\mathcal{L}_D(\phi) = \mathbb{E}_{y \sim \mathcal{D}_{\text{pos}}} \left[\frac{1}{T} \sum_{t=1}^T m_t \log D_\phi(y_t) \right] - \mathbb{E}_{y \sim \mathcal{D}_{\text{neg}}} \left[\frac{1}{T} \sum_{t=1}^T m_t \log (1 - D_\phi(y_t)) \right]. \quad (3)$$

After each critic update, we convert the probabilities into token rewards:

$$v_\phi(y_t) = \log D_\phi(y_t) - \log(1 - D_\phi(y_t)). \quad (4)$$

4.3 Policy Optimisation with Learned Rewards

Directly optimising Eq. (1) is a bilevel problem, which is computationally expensive for LLMs. We therefore alternate updates: critic steps for ϕ , then policy steps for θ under the current reward.

We optimise the LLM policy π_θ with Group Relative Policy Optimisation (GRPO) [Shao et al., 2024]. At each policy iteration, we freeze θ_{old} , sample a group of G traces $\{y^{(g)}\}_{g=1}^G \sim \pi_{\theta_{\text{old}}}$ for prompt x , and update θ while keeping θ_{old} fixed.

Reward Densification. The reward logit $v_\phi(y_t)$ is only defined at supervised positions ($m_t = 1$). For *sparse* and *interval*, we propagate each supervised score backwards to the preceding unsupervised tokens:

$$t' = \min\{k \geq t \mid m_k = 1\}, \quad r_\phi(y_t) = v_\phi(y_{t'}).$$

For *dense*, $r_\phi(y_t) = v_\phi(y_t)$. Intuitively, the latest supervised token for a segment has the richest context about the reasoning that led to it, so earlier tokens in that segment inherit its score.

Advantage Calculation. Following Cetin et al. [2025] and Cui et al. [2025]’s approach for dense rewards in GRPO, we compute length-normalised sequence-level reward means

$$\bar{r}^{(g)} = \frac{1}{T_g} \sum_{t=1}^{T_g} r_\phi(y_t^{(g)}),$$

then calculate the advantage scores by normalising within each group:

$$A^{(g)} = \frac{\bar{r}^{(g)} - \mu}{\sigma + \epsilon}. \quad (5)$$

We assign the sequence-level advantage $A^{(g)}$ to all tokens in generation g , and optimise the PPO clip loss [Schulman et al., 2017] on policy-generated traces only (regardless of whether they fall in $\mathcal{D}_{\text{pos}}$ or $\mathcal{D}_{\text{neg}}$).

Additional details on adversarial stability are provided in Appendix A; the full algorithm is given in Algorithm 1.

5 Experiments

We evaluate our learning of reasoning reward models via R-AIRL on GSM8K [Cobbe et al., 2021], MEDREASON [Wu et al., 2025], and MMLU-PRO [Wang et al., 2024]. We choose these benchmarks because they stress complementary reasoning regimes: GSM8K emphasises multi-step arithmetic computation, with demonstrations collected from humans. MEDREASON emphasises domain-specific clinical reasoning with plausible but incorrect alternatives, with demonstrations generated by LLMs and quality checked by medical doctors. MMLU-PRO emphasises broad, heterogeneous scientific reasoning, and the reasoning demonstrations are generated by LLMs. This diversity is important for assessing whether the learned reward remains useful across different reasoning styles, rather than a single-task format.

Our base policies are open-weight instruction-tuned variants not trained for reasoning: Qwen2.5 (7B) [Bai et al., 2023], Llama3 (8B) [Touvron et al., 2023], and Qwen3-4B [Yang et al., 2025]. We always instantiate a policy model as a text generator and the reasoning reward model as a token classifier.

Each experimental subsection directly evaluates one desideratum from Section 1. [D1](#) (Section 5.1) tests training-signal quality in the demonstration-only (no external verifier) regime via final policy after R-AIRL training. [D2](#) (Section 5.2) tests inference-time utility via reranking gains under fixed budgets and transfer across tasks/backbones. [D3](#) (Section 5.3) tests process-level utility via localisation of the erroneous reasoning step. Implementation details appear in Appendix B.

5.1 Using the Learned Reward as a Training Signal

RQ1: Can R-AIRL improve reasoning in an LLM policy π_θ (the generator) when π_θ is trained only with rewards inferred from expert demonstrations?

Experimental setup. For each backbone, we train π_θ with R-AIRL using *sparse*, *interval* ($k = 15$), and *dense* rewards. Following Ye et al. [2025], Lee et al. [2025], we use a 250-step discriminator warm-up, then jointly optimise policy and reward models per dataset. We report held-out pass@1 and compare to SFT under the same demonstration-only assumption (Table 2).

Method	Granularity	GSM8K			MMLU-PRO			MEDREASON		
		Qwen2.5-7B	Llama3.1-8B	Qwen3-4B	Qwen2.5-7B	Llama3.1-8B	Qwen3-4B	Qwen2.5-7B	Llama3.1-8B	Qwen3-4B
SFT		70.1 $\pm$ 1.6	66.6 $\pm$ 2.1	76.6 $\pm$ 1.7	48.1 $\pm$ 1.9	47.2 $\pm$ 1.9	53.9 $\pm$ 2.0	65.4 $\pm$ 2.3	79.5 $\pm$ 2.0	67.4 $\pm$ 2.3
	<i>Sparse</i>	85.8 $\pm$ 1.5	80.6 $\pm$ 1.6	90.4 $\pm$ 1.5	48.5 $\pm$ 2.2	<u>43.3</u> $\pm$ 1.9	55.6 $\pm$ 2.2	67.1 $\pm$ 2.5	<u>77.6</u> $\pm$ 2.0	68.9 $\pm$ 2.6
	<i>Interval</i>	<u>78.8</u> $\pm$ 1.6	<u>67.6</u> $\pm$ 1.9	87.8 $\pm$ 1.6	50.6 $\pm$ 1.9	36.6 $\pm$ 1.8	53.5 $\pm$ 2.4	<u>66.3</u> $\pm$ 2.5	76.3 $\pm$ 2.1	<u>69.1</u> $\pm$ 2.6
	<i>Dense</i>	38.4 $\pm$ 2.0	64.6 $\pm$ 2.1	<u>89.6</u> $\pm$ 1.5	43.8 $\pm$ 2.0	37.9 $\pm$ 1.8	<u>55.1</u> $\pm$ 2.3	<u>66.5</u> $\pm$ 2.6	75.1 $\pm$ 2.1	69.2 $\pm$ 2.5

Table 2: **Held-out pass@1 in the demonstration-only setting.** Bold/underlined mark best/second-best across SFT and R-AIRL variants. Values are mean $\pm$ half-width of the bootstrapped 95% confidence interval.

Training-signal quality. Table 2 shows learned rewards are competitive but not uniformly better than SFT: selecting the best granularity per backbone–dataset pair, R-AIRL improves over SFT in 7/9 settings. Gains are strongest on GSM8K (3/3 backbones) and positive for 2/3 backbones on MEDREASON; the main underperforming cases are Llama3.1-8B on MMLU-PRO and MEDREASON. Granularity controls stability: *sparse* is best in 7/9 pairs, *interval* in 1/9, and *dense* in 1/9. RQ1 targets only [D1](#); even where SFT has higher pass@1, it cannot provide the learned reward needed for [D2](#) reranking and [D3](#) process localisation (Sections 5.2 and 5.3).

Training dynamics. Figure 2 (Qwen2.5-7B on MMLU-PRO) should be read jointly: learned reward (left), discriminator accuracy (middle), and held-out pass@1 (right). In R-AIRL, discriminator accuracy approaching 50% is meaningful only if pass@1 rises; rising reward (2a) or falling discriminator accuracy (2b) without sustained pass@1 gains (2c) indicates reward–task decoupling, more common under denser supervision. We attribute *dense* instability to token-level supervision, which amplifies critic misclassifications and yields noisier policy rewards; *sparse* rewards are less sensitive and thus more stable.

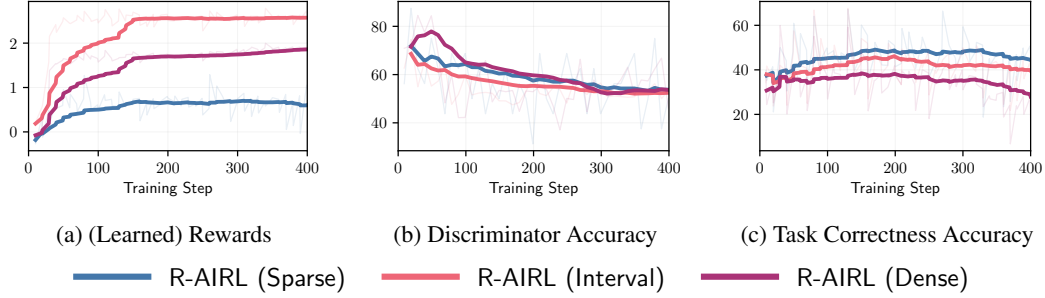


Figure 2: **Training dynamics for Qwen2.5-7B on MMLU-PRO.** Left: learned reward; middle: discriminator accuracy; right: held-out pass@1 (diagnostic only; not used for training updates). Reward improvement without pass@1 gains indicates reward–task decoupling.

For completeness, Appendix Table 10 reports a comparison against outcome rewards if external verifiers were available. Appendix Figures 6–8 show full training curves. Moreover, Appendix C presents ablations on expert corruptions, group size, clipping, and policy-vs-expert-only separation, as well as on the AIME 2024/2025 datasets.

Takeaway. RQ1 is answered positively: in the demonstration-only setting, learned rewards provide a viable training signal, with a granularity trade-off between optimisation stability (sparser rewards) and richer supervision (denser rewards).

5.2 Inference-time Assistance via Reward-guided Reranking

RQ2: *Can the learned reasoning reward improve inference by reranking multiple sampled traces under a fixed test-time budget?*

Experimental setup. We evaluate inference-time reranking with the learned R-AIRL reward: for each prompt, sample $N = 16$ traces, rank by mean reward, and return the top candidate (Best-of-16). We compare against pass@1 with random selection from the same sample set. Main results are in Table 3, and calibration (AUROC/ECE) is in Appendix Table 11.

Method	Granularity	GSM8K			MMLU-PRO			MEDREASON		
		Random	Reward	Δ	Random	Reward	Δ	Random	Reward	Δ
Qwen2.5-7B	Sparse	85.8 $\pm$ 1.5	88.8 $\pm$ 1.7	($\uparrow$ +3.0)	48.5 $\pm$ 2.2	51.5 $\pm$ 2.5	($\uparrow$ +3.0)	67.1 $\pm$ 2.5	70.1 $\pm$ 2.9	($\uparrow$ +3.0)
	Interval	78.8 $\pm$ 1.6	82.5 $\pm$ 2.0	($\uparrow$ +3.7)	50.6 $\pm$ 1.9	53.1 $\pm$ 2.5	($\uparrow$ +2.5)	66.3 $\pm$ 2.5	67.2 $\pm$ 2.9	($\uparrow$ +0.9)
	Dense	38.4 $\pm$ 2.0	55.7 $\pm$ 2.6	($\uparrow$ +17.4)	43.8 $\pm$ 2.0	46.0 $\pm$ 2.5	($\uparrow$ +2.2)	66.5 $\pm$ 2.6	66.6 $\pm$ 2.9	($\uparrow$ +0.1)
Llama3.1-8B	Sparse	80.6 $\pm$ 1.6	82.1 $\pm$ 2.1	($\uparrow$ +1.5)	43.3 $\pm$ 1.9	42.3 $\pm$ 2.4	($\downarrow$ -1.0)	77.6 $\pm$ 2.0	79.9 $\pm$ 2.4	($\uparrow$ +2.3)
	Interval	67.6 $\pm$ 1.9	78.2 $\pm$ 2.2	($\uparrow$ +10.7)	36.6 $\pm$ 1.8	38.1 $\pm$ 2.4	($\uparrow$ +1.6)	76.3 $\pm$ 2.1	81.4 $\pm$ 2.5	($\uparrow$ +5.1)
	Dense	64.6 $\pm$ 2.1	71.1 $\pm$ 2.4	($\uparrow$ +6.5)	37.9 $\pm$ 1.8	39.7 $\pm$ 2.5	($\uparrow$ +1.9)	75.1 $\pm$ 2.1	80.9 $\pm$ 2.4	($\uparrow$ +5.8)
Qwen3-4B	Sparse	90.4 $\pm$ 1.5	93.0 $\pm$ 1.4	($\uparrow$ +2.5)	55.6 $\pm$ 2.2	62.1 $\pm$ 2.4	($\uparrow$ +6.4)	68.9 $\pm$ 2.6	69.9 $\pm$ 2.9	($\uparrow$ +1.0)
	Interval	87.8 $\pm$ 1.6	91.2 $\pm$ 1.5	($\uparrow$ +3.5)	53.5 $\pm$ 2.4	62.2 $\pm$ 2.4	($\uparrow$ +8.7)	69.1 $\pm$ 2.6	71.4 $\pm$ 2.9	($\uparrow$ +2.3)
	Dense	89.6 $\pm$ 1.5	91.4 $\pm$ 1.5	($\uparrow$ +1.8)	55.1 $\pm$ 2.3	59.2 $\pm$ 2.4	($\uparrow$ +4.1)	69.2 $\pm$ 2.5	69.9 $\pm$ 2.8	($\uparrow$ +0.7)

Table 3: **Best-of-16 reranking performance (%)**. Reward-guided reranking is compared against random selection from the same sample set. Δ reports percentage-point change.

Reranking performance. In Table 3, reward-guided reranking improves pass@1 in 26/27 settings at $N = 16$. Largest gains include +17.4 pp (Qwen2.5-7B+dense, GSM8K), +10.7 pp (Llama3.1-8B+interval, GSM8K), and +8.7 pp (Qwen3-4B+interval, MMLU-PRO). On MEDREASON, gains are smaller but positive in all 9 settings (+0.1 to +5.8 pp). The only drop is Llama3.1-8B+sparse on MMLU-PRO (−1.0 pp).

Transfer across tasks and backbones. Using *interval* reward models trained on different dataset-/backbones to rerank natural Qwen2.5-7B SFT samples, Table 4 shows uniformly positive transfer: all 18 source–target pairs improve, including off-diagonal gains up to +12.1 pp (Llama3.1-8B) and +13.0 pp (Qwen3-4B).

Policy Dataset	Llama3.1-8B			Qwen3-4B		
	GSM8K	MMLU-PRO	MEDREASON	GSM8K	MMLU-PRO	MEDREASON
GSM8K	$\uparrow +12.1$	$\uparrow +7.7$	$\uparrow +12.1$	$\uparrow +13.0$	$\uparrow +8.8$	$\uparrow +13.0$
MMLU-PRO	$\uparrow +4.9$	$\uparrow +4.4$	$\uparrow +4.4$	$\uparrow +9.3$	$\uparrow +1.9$	$\uparrow +1.9$
MEDREASON	$\uparrow +3.0$	$\uparrow +10.1$	$\uparrow +11.5$	$\uparrow +5.5$	$\uparrow +5.6$	$\uparrow +5.8$

Table 4: **Best-of-16 transfer reranking deltas (%)**. Each cell reports Δ (reward-guided minus random reranking on the same sampled generations). Columns group reward-model backbones; rows are policy-task datasets. Diagonal entries (italic) are in-distribution.

Robustness across rerankers and budgets. Across $N \in \{2, 3, 5, 8, 16\}$, we compare log-probability, reward-guided reranking, majority voting, and reward-weighted majority voting on SFT-generated candidates (so logits remain independent of the reward model). Figure 3a shows a consistent trend: reward-guided reranking is strongest at low N , majority voting improves with N , and reward-weighted majority voting performs best overall. Appendix Tables 12–16 show the same pattern.

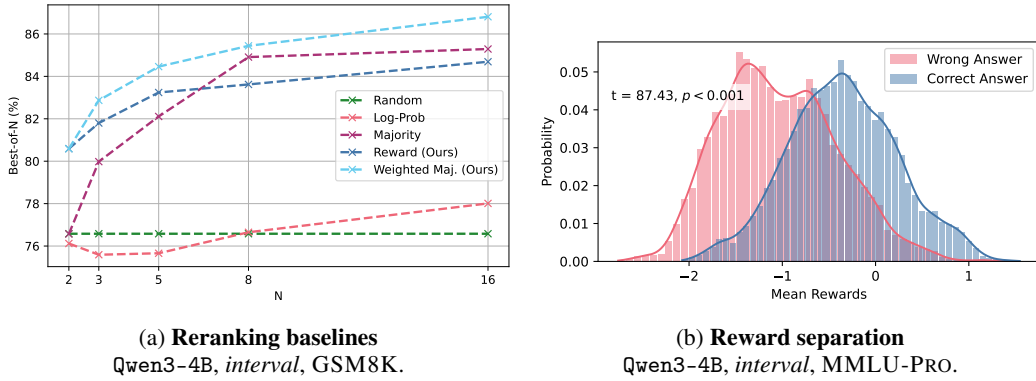


Figure 3: **Inference-time behaviour of learned rewards.** 3a: low-budget gains from reward-guided reranking; at higher N , reward-weighted majority voting is strongest. 3b: correct traces receive higher rewards than incorrect traces ($t = 87.43, p < 0.001$).

Why reranking works. Figure 3b shows clear reward separation between correct and incorrect traces ($t = 87.43, p < 0.001$) for Qwen3-4B+interval on MMLU-PRO, explaining the reranking gains in Table 3. Dataset-wide separation plots are in Appendix Figures 9–17.



Takeaway. RQ2 is answered positively: the learned reward is a reusable inference-time ranking signal that transfers across settings and complements consistency-based voting under fixed budgets.

5.3 Reasoning Error Localisation

RQ3: *Do the learned reasoning rewards provide process-level feedback that localises where reasoning fails?*

Experimental setup. To assess process-level supervision, we test whether dense rewards identify where reasoning first fails on GSM8K, and complement this with qualitative paired traces in Figure 4. Starting from correctly answered Qwen2.5-7B SFT traces, we construct controlled examples with a single perturbation in the reasoning chain (operator swaps or number corruption) while keeping the final answer. We score perturbed traces with our reasoning reward models under *dense* and *interval* granularities. For each corrupted trace, let r_t denote the reward at token t . We estimate the error location by detecting the largest one-step decrease in reward along the trace: $\Delta_t = \max(0, r_{t-1} - r_t)$, and $\hat{t} = \arg \max_t \Delta_t$. Intuitively, $\hat{t}$ is the position where the reward signal drops most sharply. We count a hit if $\hat{t}$ lies within $\pm\{1, 7\}$ tokens of the known perturbation index (Hit@1 and Hit@7).

Localisation performance. Table 5 shows strong localisation across all backbones, with best scores of 65.5 (Qwen2.5-7B), 86.1 (Llama3.1-8B), and 78.9 (Qwen3-4B). All scores are significantly above the empirical chance, indicating a better-than-random chance of scoring the reward drop within ± 1 and ± 7 tokens of the failure cases. In Appendix D.6, we demonstrate examples of the diverging reasoning trajectories on correct and corrupted traces after the first mistake.

Error localisation on natural traces. Figure 4 shows the similar behaviour qualitatively, on correctly vs incorrectly generated traces: rewards stay positive on locally correct intermediate steps and drop sharply at the first erroneous step (here at “ $12 - 2/3(12) = 4$ ”). This suggests the reward is sensitive to where reasoning first diverges. Additional examples on all datasets are in Appendix Figures 19-24.

Model	Dense		Interval
	Hit@1	Hit@7	Hit@7
Qwen2.5-7B	47.6 ± 2.8 ($\uparrow +36.2$)	54.9 ± 2.6 ($\uparrow +23.6$)	65.5 ± 2.6 ($\uparrow +13.2$)
Llama3.1-8B	65.8 ± 2.8 ($\uparrow +53.7$)	80.6 ± 2.2 ($\uparrow +47.5$)	86.1 ± 2.0 ($\uparrow +30.1$)
Qwen3-4B	62.0 ± 2.7 ($\uparrow +50.3$)	70.3 ± 2.5 ($\uparrow +38.5$)	78.9 ± 2.2 ($\uparrow +26.0$)

Table 5: **Error Localisation.** Hit@ k (%), mean $\pm$ 95% CI half-width). ($\uparrow +\Delta pp$) denotes gain over chance. *Interval* uses 15-token buckets, so Hit@1 = Hit@7.

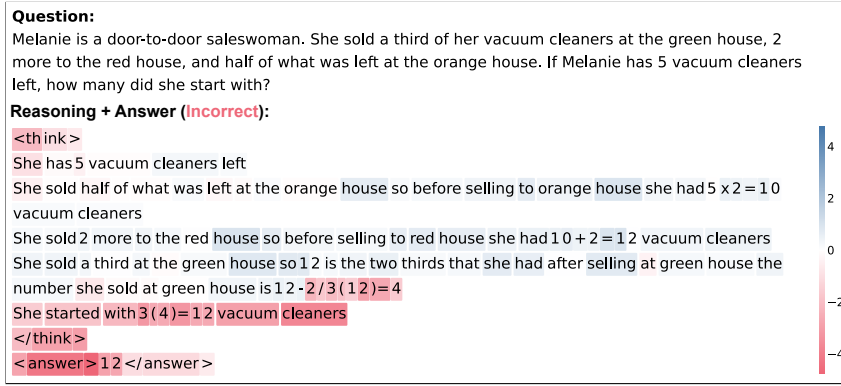


Figure 4: **Error localisation of incorrect reasoning.** Dense reward on incorrect generation for a GSM8K test question, using the *dense* Qwen2.5-7B reasoning reward model. The reward remains positive on valid intermediate steps, then drops sharply at the first erroneous step, followed by propagated penalties on later tokens.



Takeaway. RQ3 is answered positively: learned process-level rewards provide interpretable diagnostics that localise where reasoning first diverges, beyond final-answer evaluation.

6 Limitations

Our approach has three main limitations. Reward density trades off with optimisation stability: sparse step-wise signals are generally robust, whereas dense signals are more prone to adversarial mode collapse, non-stationarity, and occasional reward–correctness decoupling. R-AIRL is also computationally heavier due to alternating adversarial updates, and our evaluation is limited to arithmetic, medical, and scientific reasoning rather than open-ended or long-context tasks.

7 Conclusion

We presented Reasoning Adversarial Inverse Reinforcement Learning (R-AIRL), a framework for learning reasoning rewards from expert demonstrations without requiring external verifiers. Instead of directly imitating expert trajectories, R-AIRL learns a reusable reward model across the full pipeline. Empirically, this reward is useful in three ways: as a training signal (D1), where R-AIRL outperforms SFT in 7/9 demonstration-only settings; as an inference-time reranker (D2), where reward-guided reranking improves pass@1 in 26/27 settings at $N = 16$, with gains up to +17.4 points and uniformly positive transfer across all 18 source–target pairs; and as process-level feedback (D3), where it localises reasoning errors with Hit@7 up to 86.1% (30.1% above chance). Remaining limitations include the trade-off between reward granularity and optimisation stability, which we leave as an important direction for future work.

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References

- Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Peiyi Wang, Qihao Zhu, Runxin Xu, Ruoyu Zhang, Shirong Ma, Xiao Bi, Xiaokang Zhang, Xingkai Yu, Yu Wu, Z. F. Wu, Zhibin Gou, Zhihong Shao, Zhuoshu Li, Ziyi Gao, Aixin Liu, Bing Xue, Bingxuan Wang, Bochao Wu, Bei Feng, Chengda Lu, Chenggang Zhao, Chengqi Deng, Chong Ruan, Damai Dai, Deli Chen, Dongjie Ji, Erhang Li, Fangyun Lin, Fucong Dai, Fuli Luo, Guangbo Hao, Guanting Chen, Guowei Li, H. Zhang, Hanwei Xu, Honghui Ding, Huazuo Gao, Hui Qu, Hui Li, Jianzhong Guo, Jiashi Li, Jingchang Chen, Jingyang Yuan, Jinhao Tu, Junjie Qiu, Junlong Li, J. L. Cai, Jiaqi Ni, Jian Liang, Jin Chen, Kai Dong, Kai Hu, Kaichao You, Kaige Gao, Kang Guan, Kexin Huang, Kuai Yu, Lean Wang, Lecong Zhang, Liang Zhao, Litong Wang, Liyue Zhang, Lei Xu, Leyi Xia, Mingchuan Zhang, Minghua Zhang, Minghui Tang, Mingxu Zhou, Meng Li, Miaojun Wang, Mingming Li, Ning Tian, Panpan Huang, Peng Zhang, Qiancheng Wang, Qinyu Chen, Qiushi Du, Ruiqi Ge, Ruisong Zhang, Ruizhe Pan, Runji Wang, R. J. Chen, R. L. Jin, Ruyi Chen, Shanghao Lu, Shangyan Zhou, Shanhuang Chen, Shengfeng Ye, Shiyu Wang, Shuiping Yu, Shunfeng Zhou, Shutong Pan, S. S. Li, Shuang Zhou, Shaoqing Wu, Tao Yun, Tian Pei, Tianyu Sun, T. Wang, Wangding Zeng, Wen Liu, Wenfeng Liang, Wenjun Gao, Wenqin Yu, Wentao Zhang, W. L. Xiao, Wei An, Xiaodong Liu, Xiaohan Wang, Xiaokang Chen, Xiaotao Nie, Xin Cheng, Xin Liu, Xin Xie, Xingchao Liu, Xinyu Yang, Xinyuan Li, Xuecheng Su, Xuheng Lin, X. Q. Li, Xiangyue Jin, Xiaojin Shen, Xiaosha Chen, Xiaowen Sun, Xiaoxiang Wang, Xinnan Song, Xinyi Zhou, Xianzu Wang, Xinxia Shan, Y. K. Li, Y. Q. Wang, Y. X. Wei, Yang Zhang, Yanhong Xu, Yao Li, Yao Zhao, Yaofeng Sun, Yaohui Wang, Yi Yu, Yichao Zhang, Yifan Shi, Yiliang Xiong, Ying He, Yishi Piao, Yisong Wang, Yixuan Tan, Yiyang Ma, Yiyuan Liu, Yongqiang Guo, Yuan Ou, Yuduan Wang, Yue Gong, Yuheng Zou, Yujia He, Yunfan Xiong, Yuxiang Luo, Yuxiang You, Yuxuan Liu, Yuyang Zhou, Y. X. Zhu, Yanping Huang, Yaohui Li, Yi Zheng, Yuchen Zhu, Yunxian Ma, Ying Tang, Yukun Zha, Yuting Yan, Z. Z. Ren, Zehui Ren, Zhangli Sha, Zhe Fu, Zhean Xu, Zhenda Xie, Zhengyan Zhang, Zhewen Hao, Zhicheng Ma, Zhigang Yan, Zhiyu Wu, Zihui Gu, Zijia Zhu, Zijun Liu, Zilin Li, Ziwei Xie, Ziyang Song, Zizheng Pan, Zhen Huang, Zhipeng Xu, Zhongyu Zhang, and Zhen Zhang. Deepseek-r1 incentivizes reasoning in llms through reinforcement learning. *Nature*, 645(8081):633–638, 2025. ISSN 1476-4687. doi: 10.1038/s41586-025-09422-z. URL <http://dx.doi.org/10.1038/s41586-025-09422-z>.
- Jonathan Uesato, Nate Kushman, Ramana Kumar, Francis Song, Noah Siegel, Lisa Wang, Antonia Creswell, Geoffrey Irving, and Irina Higgins. Solving math word problems with process- and outcome-based feedback, November 2022. URL <http://arxiv.org/abs/2211.14275>. arXiv:2211.14275 [cs].
- Hunter Lightman, Vineet Kosaraju, Yura Burda, Harri Edwards, Bowen Baker, Teddy Lee, Jan Leike, John Schulman, Ilya Sutskever, and Karl Cobbe. Let’s Verify Step by Step, May 2023. URL <http://arxiv.org/abs/2305.20050>. arXiv:2305.20050 [cs].
- Amrith Setlur, Nived Rajaraman, Sergey Levine, and Aviral Kumar. Scaling Test-Time Compute Without Verification or RL is Suboptimal, February 2025. URL <http://arxiv.org/abs/2502.12118>. arXiv:2502.12118 [cs].
- Pieter Abbeel and Andrew Y. Ng. Apprenticeship learning via inverse reinforcement learning. In *Twenty-first international conference on Machine learning - ICML ’04*, page 1, Banff, Alberta, Canada, 2004. ACM Press. doi: 10.1145/1015330.1015430. URL <http://portal.acm.org/citation.cfm?doid=1015330.1015430>.
- Brian D Ziebart, Andrew Maas, J Andrew Bagnell, and Anind K Dey. Maximum Entropy Inverse Reinforcement Learning. 2004.

- Jonathan Ho and Stefano Ermon. Generative Adversarial Imitation Learning, June 2016. URL <http://arxiv.org/abs/1606.03476>. arXiv:1606.03476 [cs].
- Justin Fu, Katie Luo, and Sergey Levine. Learning Robust Rewards with Adversarial Inverse Reinforcement Learning, August 2018. URL <http://arxiv.org/abs/1710.11248>. arXiv:1710.11248 [cs].
- Qiyang Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan, Xiaochen Zuo, Yu Yue, Weinan Dai, Tiantian Fan, Gaohong Liu, Lingjun Liu, Xin Liu, Haibin Lin, Zhiqi Lin, Bole Ma, Guangming Sheng, Yuxuan Tong, Chi Zhang, Mofan Zhang, Wang Zhang, Hang Zhu, Jinhua Zhu, Jiaze Chen, Jiangjie Chen, Chengyi Wang, Hongli Yu, Yuxuan Song, Xiangpeng Wei, Hao Zhou, Jingjing Liu, Wei-Ying Ma, Ya-Qin Zhang, Lin Yan, Mu Qiao, Yonghui Wu, and Mingxuan Wang. DAPO: An Open-Source LLM Reinforcement Learning System at Scale, May 2025. URL <http://arxiv.org/abs/2503.14476>. arXiv:2503.14476 [cs].
- YuYue, Yufeng Yuan, Qiyang Yu, Xiaochen Zuo, Ruofei Zhu, Wenyuan Xu, Jiaze Chen, Chengyi Wang, TianTian Fan, Zhengyin Du, Xiangpeng Wei, Gaohong Liu, Juncai Liu, Lingjun Liu, Haibin Lin, Zhiqi Lin, Bole Ma, Chi Zhang, Mofan Zhang, Wang Zhang, Hang Zhu, Ru Zhang, Xin Liu, Mingxuan Wang, Yonghui Wu, and Lin Yan. VAPO: Efficient and Reliable Reinforcement Learning for Advanced Reasoning Tasks, April 2025. URL <http://arxiv.org/abs/2504.05118>. arXiv:2504.05118 [cs] version: 1.
- Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah D. Goodman. STaR: Bootstrapping Reasoning With Reasoning, May 2022. URL <http://arxiv.org/abs/2203.14465>. arXiv:2203.14465 [cs].
- Zheng Yuan, Hongyi Yuan, Chengpeng Li, Guanting Dong, Keming Lu, Chuanqi Tan, Chang Zhou, and Jingren Zhou. Scaling Relationship on Learning Mathematical Reasoning with Large Language Models, September 2023. URL <http://arxiv.org/abs/2308.01825>. arXiv:2308.01825 [cs].
- Avi Singh, John D. Co-Reyes, Rishabh Agarwal, Ankesh Anand, Piyush Patil, Xavier Garcia, Peter J. Liu, James Harrison, Jaehoon Lee, Kelvin Xu, Aaron T. Parisi, Abhishek Kumar, Alexander A. Alemi, Alex Rizkowsky, Azade Nova, Ben Adlam, Bernd Bohnet, Gamaleldin Fathy Elsayed, Hanie Sedghi, Igor Mordatch, Isabelle Simpson, Izzeddin Gur, Jasper Snoek, Jeffrey Pennington, Jiri Hron, Kathleen Kenealy, Kevin Swersky, Kshiteej Mahajan, Laura A. Culp, Lechao Xiao, Maxwell Bileschi, Noah Constant, Roman Novak, Rosanne Liu, Tris Warkentin, Yamini Bansal, Ethan Dyer, Behnam Neyshabur, Jascha Sohl-Dickstein, and Noah Fiedel. Beyond Human Data: Scaling Self-Training for Problem-Solving with Language Models. *Transactions on Machine Learning Research*, January 2024. ISSN 2835-8856. URL <https://openreview.net/forum?id=1NAyUngGFK>.
- Arian Hosseini, Xingdi Yuan, Nikolay Malkin, Aaron Courville, Alessandro Sordoni, and Rishabh Agarwal. V-STaR: Training Verifiers for Self-Taught Reasoners. August 2024. URL <https://openreview.net/forum?id=stmqBSW2dV#discussion>.
- David Silver, Julian Schrittwieser, Karen Simonyan, Ioannis Antonoglou, Aja Huang, Arthur Guez, Thomas Hubert, Lucas baker, Matthew Lai, Adrian Bolton, Yutian Chen, Timothy P. Lillicrap, Fan Hui, L. Sifre, George van den Driessche, Thore Graepel, and Demis Hassabis. Mastering the game of go without human knowledge. *Nature*, 550:354–359, 2017. URL <https://api.semanticscholar.org/CorpusID:205261034>.
- Paul Christiano, Jan Leike, Tom B. Brown, Miljan Martic, Shane Legg, and Dario Amodei. Deep reinforcement learning from human preferences, February 2023. URL <http://arxiv.org/abs/1706.03741>. arXiv:1706.03741.
- Rafael Rafailov, Archit Sharma, Eric Mitchell, Stefano Ermon, Christopher D. Manning, and Chelsea Finn. Direct Preference Optimization: Your Language Model is Secretly a Reward Model, July 2024. URL <http://arxiv.org/abs/2305.18290>. arXiv:2305.18290.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the Knowledge in a Neural Network, March 2015. URL <http://arxiv.org/abs/1503.02531>. arXiv:1503.02531 [stat].
- Minki Kang, Seanie Lee, Jinheon Baek, Kenji Kawaguchi, and Sung Ju Hwang. Knowledge-Augmented Reasoning Distillation for Small Language Models in Knowledge-Intensive Tasks. In A. Oh, T. Naumann, A. Globerson, K. Saenko, M. Hardt, and S. Levine, editors, *Advances in*

- Neural Information Processing Systems*, volume 36, pages 48573–48602. Curran Associates, Inc., 2023. URL https://proceedings.neurips.cc/paper_files/paper/2023/file/97faedc90260eae5c400f92d5831c3d7-Paper-Conference.pdf.
- Kalle Kujanpää, Pekka Marttinen, Harri Valpola, and Alexander Ilin. Efficient Knowledge Injection in LLMs via Self-Distillation, August 2025. URL <http://arxiv.org/abs/2412.14964>. arXiv:2412.14964 [cs].
- Hongling Xu, Qi Zhu, Heyuan Deng, Jinpeng Li, Lu Hou, Yasheng Wang, Lifeng Shang, Ruifeng Xu, and Fei Mi. KDRL: Post-Training Reasoning LLMs via Unified Knowledge Distillation and Reinforcement Learning, June 2025. URL <http://arxiv.org/abs/2506.02208>. arXiv:2506.02208 [cs].
- Idan Shenfeld, Mehul Damani, Jonas Hübötter, and Pulkit Agrawal. Self-distillation enables continual learning. In *ICLR 2026 Workshop on Lifelong Agents: Learning, Aligning, Evolving*, 2026. URL <https://openreview.net/forum?id=HlWA3V6iKF>.
- Tianzhu Ye, Li Dong, Zewen Chi, Xun Wu, Shaohan Huang, and Furu Wei. Black-box on-policy distillation of large language models. *arXiv preprint arXiv:2511.10643*, 2025. URL <https://arxiv.org/abs/2511.10643>.
- Byung-Kwan Lee, Ryo Hachiuma, Yong Man Ro, Yu-Chiang Frank Wang, and Yueh-Hua Wu. Unified reinforcement and imitation learning for vision-language models. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL <https://openreview.net/forum?id=7wEvjzkNXg>.
- Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. DeepSeekMath: Pushing the Limits of Mathematical Reasoning in Open Language Models, April 2024. URL <http://arxiv.org/abs/2402.03300>. arXiv:2402.03300 [cs].
- Edoardo Cetin, Tianyu Zhao, and Yujin Tang. Reinforcement learning teachers of test time scaling. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL <https://openreview.net/forum?id=tebG8q5EeK>.
- Ganqu Cui, Lifan Yuan, Zefan Wang, Hanbin Wang, Wendi Li, Bingxiang He, Yuchen Fan, Tianyu Yu, Qixin Xu, Weize Chen, Jiarui Yuan, Huayu Chen, Kaiyan Zhang, Xingtai Lv, Shuo Wang, Yuan Yao, Xu Han, Hao Peng, Yu Cheng, Zhiyuan Liu, Maosong Sun, Bowen Zhou, and Ning Ding. Process Reinforcement through Implicit Rewards, February 2025. URL <http://arxiv.org/abs/2502.01456>. arXiv:2502.01456 [cs].
- John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal Policy Optimization Algorithms, August 2017. URL <http://arxiv.org/abs/1707.06347>. arXiv:1707.06347 [cs].
- Karl Cobbe, Vineet Kosaraju, Mohammad Bavarian, Mark Chen, Heewoo Jun, Lukasz Kaiser, Matthias Plappert, Jerry Tworek, Jacob Hilton, Reiichiro Nakano, Christopher Hesse, and John Schulman. Training Verifiers to Solve Math Word Problems, November 2021. URL <http://arxiv.org/abs/2110.14168>. arXiv:2110.14168 [cs].
- Juncheng Wu, Wenlong Deng, Xingxuan Li, Sheng Liu, Taomian Mi, Yifan Peng, Ziyang Xu, Yi Liu, Hyunjin Cho, Chang-In Choi, Yihan Cao, Hui Ren, Xiang Li, Xiaoxiao Li, and Yuyin Zhou. Medreason: Eliciting factual medical reasoning steps in llms via knowledge graphs, 2025. URL <https://arxiv.org/abs/2504.00993>.
- Yubo Wang, Xueguang Ma, Ge Zhang, Yuansheng Ni, Abhranil Chandra, Shiguang Guo, Weiming Ren, Aaran Arulraj, Xuan He, Ziyang Jiang, et al. Mmlu-pro: A more robust and challenging multi-task language understanding benchmark. *arXiv preprint arXiv:2406.01574*, 2024.
- Jinze Bai, Shuai Bai, Yunfei Chu, Zeyu Cui, Kai Dang, Xiaodong Deng, Yang Fan, Wenbin Ge, Yu Han, Fei Huang, Binyuan Hui, Luo Ji, Mei Li, Junyang Lin, Runji Lin, Dayiheng Liu, Gao Liu, Chengqiang Lu, Keming Lu, Jianxin Ma, Rui Men, Xingzhang Ren, Xuancheng Ren, Chuanqi Tan, Sinan Tan, Jianhong Tu, Peng Wang, Shijie Wang, Wei Wang, Shengguang Wu, Benfeng

- Xu, Jin Xu, An Yang, Hao Yang, Jian Yang, Shusheng Yang, Yang Yao, Bowen Yu, Hongyi Yuan, Zheng Yuan, Jianwei Zhang, Xingxuan Zhang, Yichang Zhang, Zhenru Zhang, Chang Zhou, Jingren Zhou, Xiaohuan Zhou, and Tianhang Zhu. Qwen Technical Report, September 2023. URL <http://arxiv.org/abs/2309.16609>. arXiv:2309.16609 [cs].
- Hugo Touvron, Thibaut Lavril, Gautier Izacard, Xavier Martinet, Marie-Anne Lachaux, Timothée Lacroix, Baptiste Rozière, Naman Goyal, Eric Hambro, Faisal Azhar, Aurelien Rodriguez, Armand Joulin, Edouard Grave, and Guillaume Lample. LLaMA: Open and Efficient Foundation Language Models, February 2023. URL <http://arxiv.org/abs/2302.13971>. arXiv:2302.13971 [cs].
- An Yang, Anfeng Li, Baosong Yang, Beichen Zhang, Binyuan Hui, Bo Zheng, Bowen Yu, Chang Gao, Chengen Huang, Chenxu Lv, Chujie Zheng, Dayiheng Liu, Fan Zhou, Fei Huang, Feng Hu, Hao Ge, Haoran Wei, Huan Lin, Jialong Tang, Jian Yang, Jianhong Tu, Jianwei Zhang, Jianxin Yang, Jiaxi Yang, Jing Zhou, Jingren Zhou, Junyang Lin, Kai Dang, Keqin Bao, Kexin Yang, Le Yu, Lianghao Deng, Mei Li, Mingfeng Xue, Mingze Li, Pei Zhang, Peng Wang, Qin Zhu, Rui Men, Ruize Gao, Shixuan Liu, Shuang Luo, Tianhao Li, Tianyi Tang, Wenbiao Yin, Xingzhang Ren, Xinyu Wang, Xinyu Zhang, Xuancheng Ren, Yang Fan, Yang Su, Yichang Zhang, Yinger Zhang, Yu Wan, Yuqiong Liu, Zekun Wang, Zeyu Cui, Zhenru Zhang, Zhipeng Zhou, and Zihan Qiu. Qwen3 technical report. *arXiv preprint arXiv:2505.09388*, 2025.
- Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium, 2018. URL <https://arxiv.org/abs/1706.08500>.
- Martin Arjovsky, Soumith Chintala, and Léon Bottou. Wasserstein GAN, December 2017. URL <http://arxiv.org/abs/1701.07875>. arXiv:1701.07875 [stat].
- Di Jin, Eileen Pan, Nassim Oufattole, Wei-Hung Weng, Hanyi Fang, and Peter Szolovits. What disease does this patient have? a large-scale open domain question answering dataset from medical exams. *Applied Sciences*, 11(14):6421, 2021.
- Ankit Pal, Logesh Kumar Umapathi, and Malaikannan Sankarasubbu. Medmcqa: A large-scale multi-subject multi-choice dataset for medical domain question answering. In Gerardo Flores, George H Chen, Tom Pollard, Joyce C Ho, and Tristan Naumann, editors, *Proceedings of the Conference on Health, Inference, and Learning*, volume 174 of *Proceedings of Machine Learning Research*, pages 248–260. PMLR, 07–08 Apr 2022. URL <https://proceedings.mlr.press/v174/pal22a.html>.
- Guido Van Rossum and Fred L Drake Jr. *Python reference manual*. Centrum voor Wiskunde en Informatica Amsterdam, 1995.
- Adam Paszke, Sam Gross, Soumith Chintala, Gregory Chanan, Edward Yang, Zachary DeVito, Zeming Lin, Alban Desmaison, Luca Antiga, and Adam Lerer. Automatic differentiation in pytorch. 2017.
- Thomas Wolf, Lysandre Debut, Victor Sanh, Julien Chaumond, Clement Delangue, Anthony Moi, Pierric Cistac, Tim Rault, Rémi Louf, Morgan Funtowicz, Joe Davison, Sam Shleifer, Patrick von Platen, Clara Ma, Yacine Jernite, Julien Plu, Canwen Xu, Teven Le Scao, Sylvain Gugger, Mariama Drame, Quentin Lhoest, and Alexander M. Rush. HuggingFace’s Transformers: State-of-the-art Natural Language Processing, July 2020. URL <http://arxiv.org/abs/1910.03771>. arXiv:1910.03771 [cs].
- Michael Han Daniel Han and Unsloth team. Unsloth, 2023. URL <http://github.com/unslothai/unsloth>.
- Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations*, 2019. URL <https://openreview.net/forum?id=Bkg6RiCqY7>.
- Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. Lora: Low-rank adaptation of large language models, 2021. URL <https://arxiv.org/abs/2106.09685>.

A Additional Method Details

A.1 Training Stability.

A known limitation of adversarial training is that a single discriminator update per policy step leaves D_ϕ under-trained relative to π_θ , making the reward signal noisy and susceptible to reward hacking [Heusel et al., 2018, Arjovsky et al., 2017, Ye et al., 2025]. To mitigate this, we use three complementary stabilisation mechanisms. First, we maintain replay buffers $\mathcal{B}_{\text{pos}}$ and $\mathcal{B}_{\text{neg}}$ that accumulate positive and negative traces across iterations; each discriminator update samples mini-batches from these buffers rather than solely from current policy rollouts, broadening the training distribution and reducing non-stationarity. Second, we perform N_{disc} discriminator gradient steps per policy update, allowing D_ϕ to track the evolving policy more closely before the next GRPO step. Third, during policy learning, we clip token rewards:

$$r_\phi(y_t) \leftarrow \text{clip}(r_\phi(y_t), -\beta, \beta), \quad \text{where } \beta > 0 \quad (6)$$

where $\beta > 0$. This bounds extreme logit-derived rewards and prevents unstable updates. Together, these mechanisms better calibrate the reward signal and reduce mode-collapse-like behaviour in adversarial training.

A.2 Algorithm

Algorithm 1 Reasoning Adversarial Inverse Reinforcement Learning (R-AIRL)

Require: Expert traces $\mathcal{D}_E$; Iterations N_{step} ; warm-up steps N_{warm} ; discriminator updates per step

```

1:  $N_{\text{disc}}$ 
2: Initialise policy  $\pi_\theta$ , discriminator  $D_\phi$ , and replay buffers  $\mathcal{B}_{\text{pos}} \leftarrow \emptyset, \mathcal{B}_{\text{neg}} \leftarrow \emptyset$ 
3: Warm-up Phase:
4: for  $j \leftarrow 1$  to  $N_{\text{warm}}$  do
5:   Sample prompts  $x \sim \mathcal{Q}$  and generate  $\mathcal{D}_P \sim \pi_{\theta_{\text{init}}}(\cdot | x)$ 
6:    $\mathcal{D}_{\text{pos}} \leftarrow \mathcal{D}_E \cup \{y \in \mathcal{D}_P \mid \mathcal{O}(y) = \mathcal{O}(y^E)\}$ 
7:    $\mathcal{D}_{\text{neg}} \leftarrow \{y \in \mathcal{D}_P \mid \mathcal{O}(y) \neq \mathcal{O}(y^E)\} \cup \{\mathcal{C}(y^E) \mid y^E \in \mathcal{D}_E\}$ 
8:   Update  $D_\phi$  to minimise Eq. (3) using  $\mathcal{D}_{\text{pos}}$  and  $\mathcal{D}_{\text{neg}}$ 
9: end for
10: Adversarial Training Phase:
11: for  $i \leftarrow 1$  to  $N_{\text{step}}$  do
12:   Sample prompts  $x \sim \mathcal{Q}$ 
13:   Generate group  $\mathcal{D}_P \leftarrow \{y^{(g)}\}_{g=1}^G \sim \pi_\theta(\cdot | x)$ 
14:   Construct training sets based on answer correctness:
15:    $\mathcal{D}_{\text{pos}} \leftarrow \mathcal{D}_E \cup \{y \in \mathcal{D}_P \mid \mathcal{O}(y) = \mathcal{O}(y^E)\}$ 
16:    $\mathcal{D}_{\text{neg}} \leftarrow \{y \in \mathcal{D}_P \mid \mathcal{O}(y) \neq \mathcal{O}(y^E)\} \cup \{\mathcal{C}(y^E) \mid y^E \in \mathcal{D}_E\}$ 
17:    $\mathcal{B}_{\text{pos}} \leftarrow \mathcal{B}_{\text{pos}} \cup \mathcal{D}_{\text{pos}}; \mathcal{B}_{\text{neg}} \leftarrow \mathcal{B}_{\text{neg}} \cup \mathcal{D}_{\text{neg}}$  ▷ Update replay buffers
18:   Reward Model Update:
19:   for  $k \leftarrow 1$  to  $N_{\text{disc}}$  do
20:     Sample mini-batches  $\tilde{\mathcal{D}}_{\text{pos}} \sim \mathcal{B}_{\text{pos}}$  and  $\tilde{\mathcal{D}}_{\text{neg}} \sim \mathcal{B}_{\text{neg}}$ 
21:     Update  $D_\phi$  to minimise Eq. (3) using  $\tilde{\mathcal{D}}_{\text{pos}}$  and  $\tilde{\mathcal{D}}_{\text{neg}}$ 
22:   end for
23:   Policy Update:
24:   Get dense rewards  $r_\phi(y_t^{(g)})$  via backfilling logits (Eq. 4 and Eq. 6)
25:   Compute advantages  $A_t^{(g)}$  via group standardisation (Eq. 5)
26:   Optimise  $\pi_\theta$  using GRPO loss on policy generations
27: end for

```

B Implementation Details

We evaluate the proposed expert reasoning approach on GSM8K [Cobbe et al., 2021], a benchmark for grade school math problems that provides final answers and human-written demonstrations. Moreover, to demonstrate the effectiveness of the proposed method in extracting a dense reasoning reward model, we performed our experiments on MEDREASON [Wu et al., 2025], and more specifically on the MEDQA [Jin et al., 2021] subset, consisting of questions from the US medical board exam, and THE MEDMCQA [Pal et al., 2022] subset, comprising questions from the entrance exam from the Indian medical school curriculum. The dataset used in our experiments comprises approximately 7’000 questions for training and 1’000 for evaluation. In addition, Wu et al. [2025] provides quality filtered medical reasoning traces constructed by strong language models (ChatGPT), which can be used for supervised fine-tuning or, in our case, adversarial inverse RL. The MMLU-PRO dataset [Wang et al., 2024] is an advanced version of the original Massive Multi-Task Language Understanding (MMLU) dataset, comprising questions that are particularly difficult to reason about across domains such as maths, biology, health, business, law, etc. Moreover, the answer options have been expanded from 4 options to 10 options to increase difficulty. This dataset provides additional insight into more challenging and open-domain reasoning tasks.

Unless otherwise noted, we use open-weight, instruction-tuned models as base policies and train a learned reward function via adversarial inverse reinforcement learning. To obtain a dense signal, we implement the discriminator as a token classifier that shares the backbone with a language model and replaces the language modelling head with a single linear layer that outputs one logit per token. The code for all our experiments can be found in https://github.com/fanconic/expert_reasoning.

All experiments are implemented in Python [Van Rossum and Drake Jr, 1995] with PyTorch [Paszke et al., 2017] and Hugging Face Transformers [Wolf et al., 2020]. We accelerate training and evaluation with UNSLOTH [Daniel Han and team, 2023]. Unless stated otherwise, we use a starting learning rate of 1×10^{-5} for the reasoning discriminator and 5×10^{-6} for the generator. The reward reasoning model is warmed up for 250 optimisation steps, and we train for 400 adversarial optimisation steps with a batch size of 16, generating $G = 8$ samples per prompt, accumulated over eight gradient steps (actual batch size + 128). During the adversarial update, we perform one policy-model update and three reasoning-reward model updates. To prevent data drift, we implemented a replay buffer that keeps track of the 50 most recent batches and samples a balanced combination of them when feeding the reward model. Both the discriminator and policy optimisers use a cosine-annealing learning rate schedule. The discriminator optimiser has a warm-up sequence of 250 steps (same as the discriminator warm-up period), while the policy optimiser is warmed up for 50 steps. We use a quantised ADAMW [Loshchilov and Hutter, 2019] optimiser.

Data and preprocessing. Prompts consist of the problem text with a short system instruction that requests step-by-step reasoning. Demonstrations are formatted as `<think> ... </think>` followed by `<answer>... </answer>` format and provide the system instructions below. Tokenisation uses the native tokeniser of each backbone. For evaluation, we decode with temperature $T = 0.5$ and $\text{top}_p = 0.95$. We validate every run after 100 steps, save the model with the best validation accuracy on correctness, for all baselines and models, and use these weights to evaluate on the test set.

System Prompt for GSM8K and MMLU-PRO

A conversation between User and Assistant. The user asks a question, and the Assistant solves it. The assistant first thinks about the reasoning process in the mind and then provides the user with the answer. The reasoning process and answer are enclosed within `<think> </think>` and `<answer> </answer>` tags, respectively, i.e., `<think> reasoning process here </think><answer> answer here </answer>`

System Prompt for MEDREASON

You are an expert medical AI. You must analyze clinical scenarios and adhere strictly to this output format:

1. Enclose your step-by-step reasoning within '`<think>`' and '`</think>`' tags. Keep your reasoning strictly <500 words.
2. Immediately after, output your final conclusion within '`<answer>`' and '`</answer>`' tags. The content must be exactly in the format: '`<LETTER>. <ANSWER_TEXT>`', where:
 - '`<LETTER>`' is one of A, B, C, D
 - '`<ANSWER_TEXT>`' is the exact option text corresponding to that letter

Example:

```
<think>
[Your step-by-step clinical reasoning]
</think>
<answer>
C. Hyperthyroidism
</answer>
"""
```

Inference time scoring. At inference time, we draw $N = 16$ samples per prompt, compute the mean polynomially discounted ($\gamma = 0.95$) reward over the reasoning tokens for each sample, and rerank by this score. We introduce this discount to reward tokens towards the end of a sequence, as they are closer to completion.

Perturbations. To improve robustness and reduce reliance on surface form, we introduce targeted perturbations during discriminator training for both expert and policy traces. For GSM8K we apply the following random operations: (i) flip arithmetic operator signs, (ii) corrupt numeric literals by small random offsets, and (iii) swap the final answer with an earlier intermediate number. Perturbed traces are labelled as non-expert. For MEDREASON, we use ChatGPT-5.2 to generate incorrect reasoning traces that lead to incorrect results, by providing in-context the correct reasoning trace and the correct answer and instructing it to corrupt it. We do the same thing for MMLU-PRO. We provide the prompt below for corruption:

Compute. Experiments are conducted on 4 A100-class GPUs using mixed-precision training. We use gradient accumulation to match effective batch sizes across the backbones. All models operate in 4-bit mode, as provided by UNSLOTH [Daniel Han and team, 2023], to improve training speed and memory efficiency. All experiments are run on a single GPU instance to enable parallelisation.

B.1 Policy and Reward Model

Policies are initialised from instruction-tuned checkpoints and trained with the learned reward signal. The following policy backbones are used:

- Llama3.1-8B-Instruct
- Llama3.2-3B-Instruct
- Qwen2.5-3B-Instruct
- Qwen2.5-7B-Instruct
- Qwen3-4B-Instruct-2507

We instantiate the same reasoning reward models as the policy models. All models are turned into LoRA Hu et al. [2021] adapters ($r = 256$, $\alpha = 512$) and PEFT is used.

System Prompt for Corrupted Dataset

```
"""
You are an expert educator creating adversarial training examples.
Below is a question, the correct reasoning, and the correct answer.

Your task:
1. Identify the incorrect options (distractors) provided in the question.
2. For EACH incorrect option, write a plausible reasoning trace that leads to
   that specific wrong answer.
3. The reasoning should sound logical but must conclude with the wrong answer.
4. Match the formatting of the correct reasoning provided.

Format your output as sequential blocks:

<block>
<think>
[Reasoning for Distractor]
</think>
<answer>
[Distractor Label and Text]
</answer>
</block>

---
Question:
{question}

Correct Reasoning (do not use):
{correct_reasoning}

Correct Answer (do not use):
{correct_answer}
"""
```

B.2 Future Work

Our findings suggest that learned reasoning reward models can effectively serve as both a training signal and an interpretable inference-time assistant. Addressing the stability-granularity trade-off is critical; this involves improving the optimisation dynamics at the discriminator-policy interface to prevent mode collapse in dense settings, potentially through alternative objectives such as the Wasserstein GAN [Arjovsky et al., 2017]. Moreover, the dense reward token-level interpretability opens the door to active test-time intervention, such as reward-guided decoding, early-exit mechanisms, and iterative self-revision strategies.

B.3 Statement about the Use of Large Language Models

We utilised large language models to assist with drafting and editing the manuscript and to accelerate implementation by generating boilerplate code and providing debugging suggestions. LLMs were not involved in the design of the methods, the study design, or the interpretation of the results. All outputs were reviewed and verified by the authors, who take full responsibility for the content.

C Additional Experiments

C.1 Ablation Reward Clipping

RQ-A1: *How sensitive is R-AIRL to reward clipping bounds once the stabilised adversarial training recipe is used?*

Experimental setup. We ablate the clipping interval $\beta = (\text{LB}, \text{UB})$ for Llama3.2-3B with step-wise rewards on GSM8K, while keeping the rest of the optimisation recipe fixed (KL penalty, replay buffer, and three discriminator updates per policy update). We report the held-out Pass@1, the reranking gain (ΔRerank), AUROC, and ECE.

β (LB, UB)	Pass@1	Δ Rerank	AUROC $\uparrow$	ECE $\downarrow$
(0, 2)	54 [52, 56]	+14	86 [85, 86]	21 [20, 22]
(-1, 1)	57 [55, 59]	+11	84 [83, 84]	15 [14, 15]
(-3, 3)	54 [52, 56]	+14	89 [88, 89]	13 [12, 13]
(-5, 5)	58 [56, 60]	+11	83 [82, 83]	20 [20, 21]
(-10, 10)	52 [51, 54]	+17	87 [87, 88]	18 [17, 18]
$(-\infty, \infty)$	58 [56, 60]	+12	87 [87, 88]	12 [12, 13]

Table 6: **Ablation on reward clipping (β)** for Llama3.2-3B (Step-wise) on GSM8K. All metrics are shown in full percentages (%). Intervals indicate 95% confidence intervals.



Takeaway. Under our stabilised adversarial recipe for R-AIRL, reward clipping is a secondary design choice rather than a primary driver of performance.

C.2 Ablation Groups

RQ-A2: *How sensitive is performance to the number of sampled generations per prompt (g) during policy optimisation and reranking?*

Experimental setup. We vary the group size $g \in \{4, 8, 16\}$ for Llama3.2-3B with step-wise rewards on GSM8K. As above, we report Pass@1, Δ Rerank, AUROC, and ECE to capture both downstream accuracy and reward quality.

Gen. (g)	Pass@1	Δ Rerank	AUROC $\uparrow$	ECE $\downarrow$
4	53 [52, 55]	+18	89 [89, 90]	13 [13, 14]
8	58 [56, 60]	+11	83 [82, 83]	20 [20, 21]
16	57 [56, 59]	+13	86 [86, 87]	14 [13, 14]

Table 7: **Ablation on number of generations (G)** for Llama3.2-3B (Interval) on GSM8K. All metrics are in full percentages (%). Intervals indicate 95% confidence intervals.



Takeaway. The method is not brittle to the exact group size, suggesting that its gains are not tied to a narrowly tuned sampling configuration.

C.3 AIME

RQ-A3: *Do demonstration-only learned rewards remain competitive on harder olympiad-style math benchmarks?*

Experimental setup. We evaluate pass@1 on AIME 2024 and AIME 2025 for the Qwen3-4B, comparing SFT with R-AIRL variants across reward granularities (sparse, interval, dense).

Method	AIME 2024	AIME 2025
Qwen3-4B		
SFT	3.45	3.12
Ours (<i>Sparse</i>)	3.43	3.82
Ours (<i>Interval</i>)	3.31	4.30
Ours (<i>Dense</i>)	3.00	2.17

Table 8: **Reward-weighted Pass@1 Performance (%)**. **Bold** indicates the best performance compared between SFT and our methods.



Takeaway. On harder olympiad-style benchmarks, the main message is competitiveness rather than uniform dominance, with stronger backbones benefiting more from reward-based training.

C.4 Ablation Corruptions

RQ-A4: *Do answer agreement and targeted corruptions reduce style shortcuts and improve reasoning-discriminative rewards?*

Experimental setup. We run a progressive ablation on Qwen2.5-3B (interval) on GSM8K: no answer agreement, answer agreement as soft supervision only, then answer agreement plus number corruption, number swapping, or symbol corruption. We report Pass@1, Δ Rerank, AUROC, and ECE (Table 9), and analyse reward correlation with correctness versus format cues (Figure 5).

Corruption Type	Pass@1	Δ Rerank	AUROC $\uparrow$	ECE $\downarrow$
No Answer Agreement	31 [29, 32]	+4	62 [61, 62]	8 [7, 8]
Answer Agreement	56 [55, 58]	+8	77 [76, 78]	22 [22, 23]
+ Number corruption	58 [56, 59]	+10	78 [77, 78]	34 [34, 35]
+ Number swapping	56 [54, 58]	+15	84 [83, 84]	27 [27, 28]
+ Symbol corruption	56 [55, 58]	+18	85 [84, 85]	31 [31, 32]

Table 9: **Ablation on Reasoning Corruption** for Qwen2.5-3B (interval) on GSM8K. All metrics are in full percentages (%). Intervals indicate 95% confidence intervals.

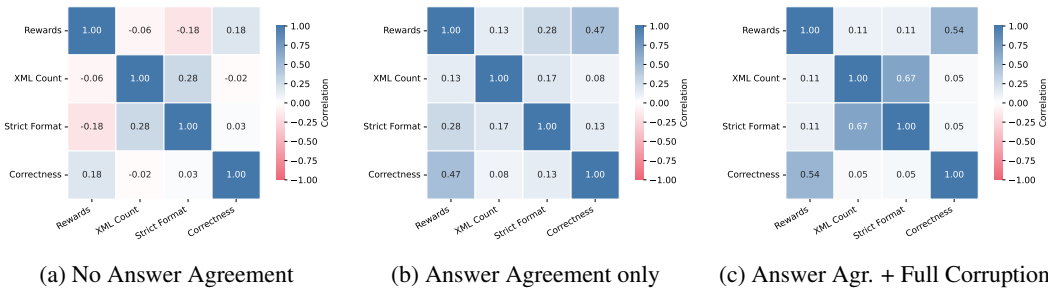


Figure 5: **Correlation of rewards with correctness and format.** Qwen2.5-3B on GSM8K under increasing corruption (no answer agreement $\rightarrow$ answer agreement $\rightarrow$ answer agreement + full corruption). Correctness correlation increases (0.18 $\rightarrow$ 0.47 $\rightarrow$ 0.54), while format correlation is reduced in the fully corrupted setting (0.28 $\rightarrow$ 0.11).



Takeaway. Answer agreement is key to learning rewards that track reasoning quality rather than superficial style cues. Corruptions can make this further robust.

D Additional Results

D.1 Complete Performance Table

Method	Granularity	GSM8K			MMLU-Pro			MEDREASON		
		Qwen2.5-7B	Llama3.1-8B	Qwen3-4B	Qwen2.5-7B	Llama3.1-8B	Qwen3-4B	Qwen2.5-7B	Llama3.1-8B	Qwen3-4B
External verifier										
Outcome Reward	–	89.0 ± 1.5	83.3 ± 1.6	91.6 ± 1.3	53.5 ± 2.0	48.4 ± 2.0	57.1 ± 2.2	69.2 ± 2.4	82.8 ± 2.1	71.8 ± 2.5
Demonstrations only										
SFT	–	70.1 ± 1.6	66.6 ± 2.1	76.6 ± 1.7	48.1 ± 1.9	47.2 ± 1.9	53.9 ± 2.0	65.4 ± 2.3	79.5 ± 2.0	67.4 ± 2.3
R-AIRL	<i>Sparse</i>	85.8 ± 1.5	80.6 ± 1.6	90.4 ± 1.5	48.5 ± 2.2	<u>43.3</u> ± 1.9	55.6 ± 2.2	<u>67.1</u> ± 2.5	<u>77.6</u> ± 2.0	68.9 ± 2.6
	<i>Interval</i>	<u>78.8</u> ± 1.6	<u>67.6</u> ± 1.9	87.8 ± 1.6	50.6 ± 1.9	36.6 ± 1.8	53.5 ± 2.4	66.3 ± 2.5	76.3 ± 2.1	<u>69.1</u> ± 2.6
	<i>Dense</i>	38.4 ± 2.0	64.6 ± 2.1	<u>89.6</u> ± 1.5	43.8 ± 2.0	37.9 ± 1.8	<u>55.1</u> ± 2.3	67.6 ± 2.4	76.7 ± 2.0	69.3 ± 2.6

Table 10: **Held-out pass@1 by supervision regime.** Methods are separated into *External verifier* (Outcome Reward) and *Demonstrations only* (SFT and R-AIRL variants). **Bold** indicates the best performance and underlined the second best among demonstration-only methods. Values are reported as mean $\pm$ half-width of the 95% confidence interval bootstrapped over the test set.

D.2 Training Behaviour

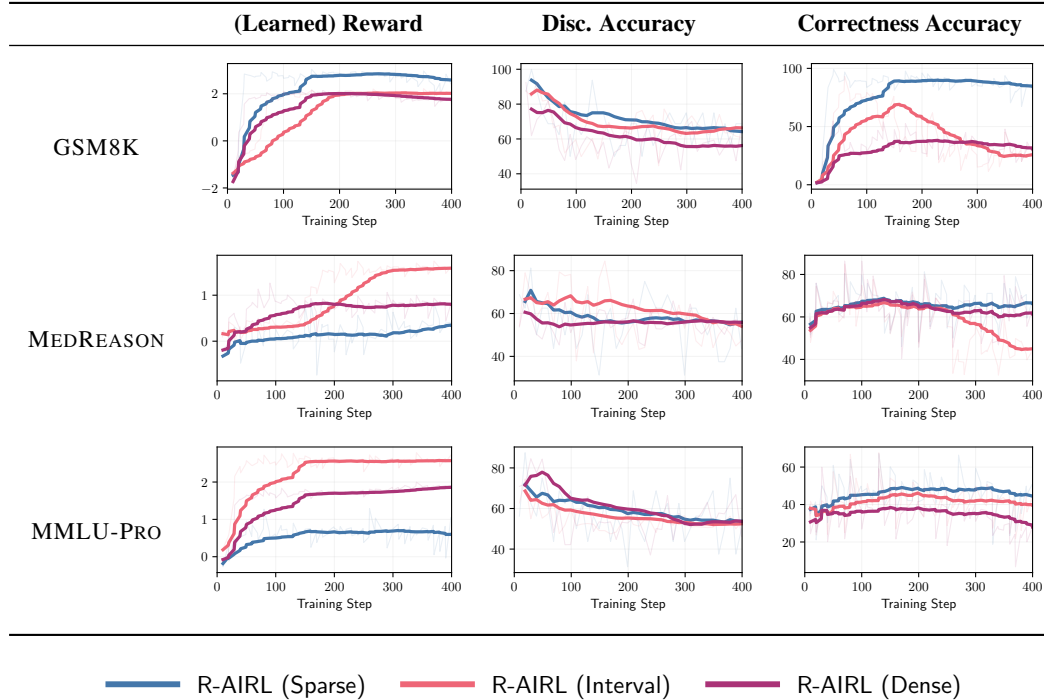


Figure 6: **Training dynamics for Qwen2.5-7B.**

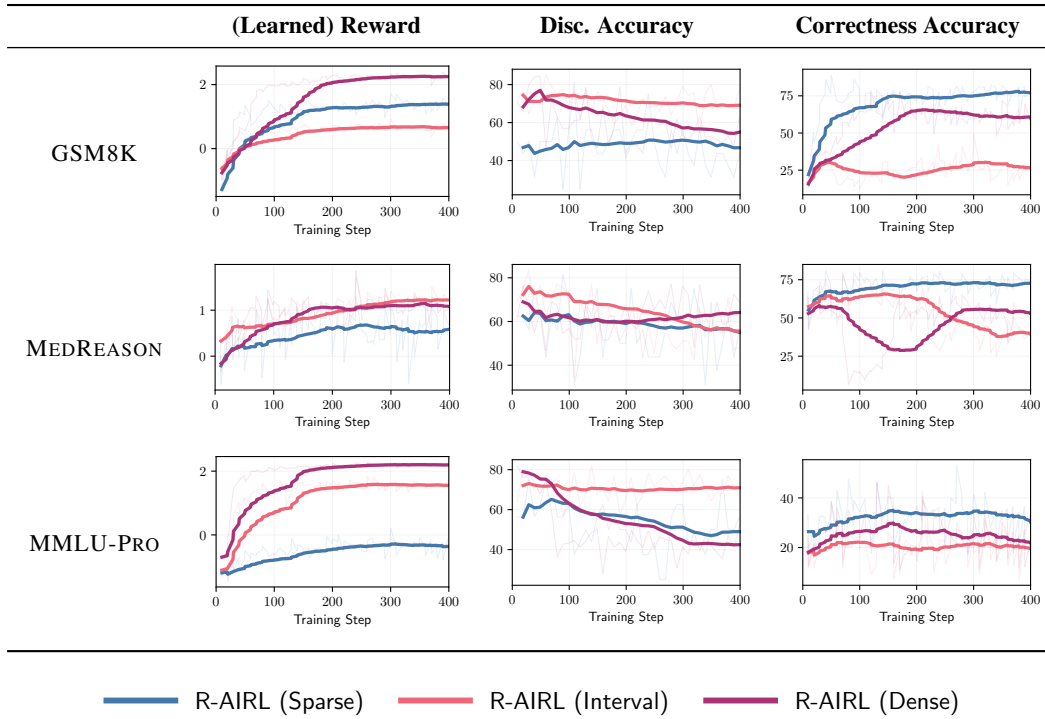


Figure 7: Training dynamics for Llama3.1-8B.

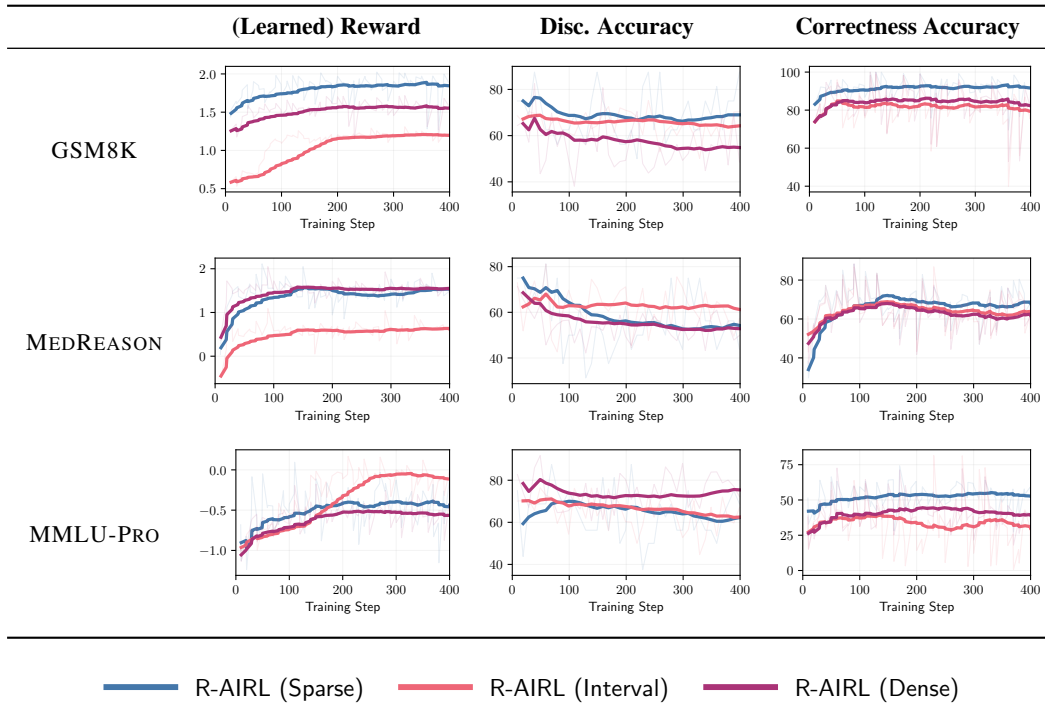


Figure 8: Training dynamics for Qwen3-4B.

D.3 Reranking Calibration

Backbone	Method	GSM8K		MEDREASON		MMLU-Pro	
		AUROC (%) ↑	ECE (%) ↓	AUROC (%) ↑	ECE (%) ↓	AUROC (%) ↑	ECE (%) ↓
Qwen2.5-7B	<i>Sparse</i>	69.9 [68.8, 71.0]	8.3 [7.9, 8.8]	76.8 [76.0, 77.6]	15.8 [15.1, 16.4]	70.9 [70.3, 71.5]	17.2 [16.6, 17.7]
	<i>Interval</i>	70.1 [69.1, 71.0]	7.5 [7.0, 8.0]	71.0 [70.2, 71.8]	9.7 [9.0, 10.4]	68.0 [67.3, 68.7]	35.6 [35.0, 36.2]
	<i>Dense</i>	71.4 [70.7, 72.1]	45.2 [44.6, 45.9]	70.4 [69.5, 71.2]	15.5 [14.8, 16.2]	60.2 [59.5, 60.9]	41.3 [40.7, 41.9]
Llama3.1-8B	<i>Sparse</i>	75.0 [74.1, 75.9]	2.4 [1.9, 2.9]	82.2 [81.5, 83.0]	14.1 [13.5, 14.6]	68.7 [68.0, 69.3]	9.8 [9.2, 10.3]
	<i>Interval</i>	81.8 [81.2, 82.5]	8.7 [8.2, 9.2]	85.1 [84.5, 85.7]	6.5 [6.0, 7.1]	70.8 [70.1, 71.5]	39.7 [39.2, 40.2]
	<i>Dense</i>	75.9 [75.1, 76.6]	27.2 [26.6, 27.8]	86.0 [85.4, 86.6]	10.9 [10.3, 11.5]	60.2 [59.4, 60.9]	50.9 [50.4, 51.5]
Qwen3-4B	<i>Sparse</i>	84.1 [83.0, 85.1]	6.8 [6.5, 7.1]	73.9 [73.1, 74.7]	11.0 [10.4, 11.7]	86.8 [86.3, 87.2]	9.9 [9.4, 10.3]
	<i>Interval</i>	78.3 [77.1, 79.5]	13.5 [13.2, 13.9]	74.8 [73.9, 75.6]	3.2 [2.6, 3.9]	84.2 [83.7, 84.7]	21.5 [21.0, 22.0]
	<i>Dense</i>	78.3 [77.2, 79.4]	7.3 [7.0, 7.7]	73.3 [72.5, 74.1]	8.1 [7.4, 8.8]	75.0 [74.4, 75.7]	18.6 [18.0, 19.2]

Table 11: **Critic Calibration Metrics.** All values are reported as percentages (%). AUROC indicates ranking ability; ECE measures calibration error (lower is better).

D.4 Reranking Performance Ablation

Backbone	Method	GSM8K					MMLU-Pro				
		Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.	Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.
Qwen2.5-4B	<i>Sparse</i>	76.6%	(↓ -0.5)	(0.0)	(↑ +3.2)	(↑ +3.2)	53.9%	(↑ +2.7)	(↑ +0.5)	(↑ +4.5)	(↑ +4.5)
	<i>Interval</i>	76.6%	(↓ -0.5)	(0.0)	(↑ +4.0)	(↑ +4.0)	53.9%	(↑ +2.7)	(↑ +0.5)	(↑ +0.9)	(↑ +0.9)
	<i>Dense</i>	76.6%	(↓ -0.5)	(0.0)	(↑ +3.6)	(↑ +3.6)	53.9%	(↑ +2.7)	(↑ +0.5)	(↑ +1.9)	(↑ +1.9)
Qwen2.5-7B	<i>Sparse</i>	70.1%	(↑ +0.4)	(↑ +0.3)	(↑ +6.0)	(↑ +6.0)	48.1%	(↑ +0.9)	(↓ -0.6)	(↑ +3.5)	(↑ +3.5)
	<i>Interval</i>	70.1%	(↑ +0.4)	(↑ +0.3)	(↑ +5.4)	(↑ +5.4)	48.1%	(↑ +0.9)	(↓ -0.6)	(↑ +1.1)	(↑ +1.1)
	<i>Dense</i>	70.1%	(↑ +0.4)	(↑ +0.3)	(↑ +6.4)	(↑ +6.4)	48.1%	(↑ +0.9)	(↓ -0.6)	(↓ -0.1)	(↓ -0.1)
Llama3.1-8B	<i>Sparse</i>	66.6%	(↑ +1.4)	(↑ +0.2)	(↑ +2.3)	(↑ +2.3)	47.2%	(↑ +1.7)	(↑ +0.6)	(↑ +2.8)	(↑ +2.8)
	<i>Interval</i>	66.6%	(↑ +1.4)	(↑ +0.2)	(↑ +3.1)	(↑ +3.1)	47.2%	(↑ +1.7)	(↑ +0.6)	(↑ +3.7)	(↑ +3.7)
	<i>Dense</i>	66.6%	(↑ +1.4)	(↑ +0.2)	(↑ +3.0)	(↑ +3.0)	47.2%	(↑ +1.7)	(↑ +0.6)	(↑ +2.0)	(↑ +2.0)

Table 12: **Best-of-2 Reranking Performance & Baselines (%)**. Random is SFT pass@1. Deltas are percentage-point changes for each reranker. **Blue arrows (↑)** indicate gains and **purple arrows (↓)** indicate drops.

Backbone	Method	GSM8K					MMLU-Pro				
		Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.	Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.
Qwen2.5-4B	<i>Sparse</i>	76.6%	(↓ -1.0)	(↑ +3.4)	(↑ +5.4)	(↑ +6.5)	53.9%	(↑ +3.7)	(↑ +3.5)	(↑ +6.5)	(↑ +6.4)
	<i>Interval</i>	76.6%	(↓ -1.0)	(↑ +3.4)	(↑ +5.2)	(↑ +6.3)	53.9%	(↑ +3.7)	(↑ +3.5)	(↑ +2.1)	(↑ +3.3)
	<i>Dense</i>	76.6%	(↓ -1.0)	(↑ +3.4)	(↑ +5.2)	(↑ +6.1)	53.9%	(↑ +3.7)	(↑ +3.5)	(↑ +3.7)	(↑ +4.3)
Qwen2.5-7B	<i>Sparse</i>	70.1%	(↑ +1.7)	(↑ +7.5)	(↑ +8.1)	(↑ +9.2)	48.1%	(↑ +1.9)	(↑ +2.1)	(↑ +4.3)	(↑ +4.5)
	<i>Interval</i>	70.1%	(↑ +1.7)	(↑ +7.5)	(↑ +6.4)	(↑ +8.5)	48.1%	(↑ +1.9)	(↑ +2.1)	(↑ +2.4)	(↑ +3.4)
	<i>Dense</i>	70.1%	(↑ +1.7)	(↑ +7.5)	(↑ +9.4)	(↑ +10.1)	48.1%	(↑ +1.9)	(↑ +2.1)	(↑ +0.2)	(↑ +2.5)
Llama3.1-8B	<i>Sparse</i>	66.6%	(↑ +2.3)	(↑ +1.8)	(↑ +2.2)	(↑ +3.4)	47.2%	(↑ +2.0)	(↑ +4.0)	(↑ +2.1)	(↑ +3.2)
	<i>Interval</i>	66.6%	(↑ +2.3)	(↑ +1.8)	(↑ +3.9)	(↑ +4.4)	47.2%	(↑ +2.0)	(↑ +4.0)	(↑ +4.0)	(↑ +4.8)
	<i>Dense</i>	66.6%	(↑ +2.3)	(↑ +1.8)	(↑ +3.7)	(↑ +3.6)	47.2%	(↑ +2.0)	(↑ +4.0)	(↑ +2.8)	(↑ +4.4)

Table 13: **Best-of-3 Reranking Performance & Baselines (%)**. Random is SFT pass@1. Deltas are percentage-point changes for each reranker. **Blue arrows (↑)** indicate gains and **purple arrows (↓)** indicate drops.

Backbone	Method	GSM8K					MMLU-Pro				
		Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.	Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.
Qwen2.5-4B	<i>Sparse</i>	76.6%	($\downarrow$ -0.9)	($\uparrow$ +5.5)	($\uparrow$ +6.8)	($\uparrow$ +7.9)	53.9%	($\uparrow$ +2.8)	($\uparrow$ +5.7)	($\uparrow$ +7.5)	($\uparrow$ +8.1)
	<i>Interval</i>	76.6%	($\downarrow$ -0.9)	($\uparrow$ +5.5)	($\uparrow$ +6.7)	($\uparrow$ +7.9)	53.9%	($\uparrow$ +2.8)	($\uparrow$ +5.7)	($\uparrow$ +2.1)	($\uparrow$ +4.8)
	<i>Dense</i>	76.6%	($\downarrow$ -0.9)	($\uparrow$ +5.5)	($\uparrow$ +6.4)	($\uparrow$ +8.1)	53.9%	($\uparrow$ +2.8)	($\uparrow$ +5.7)	($\uparrow$ +3.5)	($\uparrow$ +5.2)
Qwen2.5-7B	<i>Sparse</i>	70.1%	($\uparrow$ +2.5)	($\uparrow$ +10.9)	($\uparrow$ +9.9)	($\uparrow$ +12.3)	48.1%	($\uparrow$ +3.5)	($\uparrow$ +4.9)	($\uparrow$ +5.9)	($\uparrow$ +6.5)
	<i>Interval</i>	70.1%	($\uparrow$ +2.5)	($\uparrow$ +10.9)	($\uparrow$ +6.8)	($\uparrow$ +11.7)	48.1%	($\uparrow$ +3.5)	($\uparrow$ +4.9)	($\uparrow$ +2.1)	($\uparrow$ +4.8)
	<i>Dense</i>	70.1%	($\uparrow$ +2.5)	($\uparrow$ +10.9)	($\uparrow$ +10.2)	($\uparrow$ +12.4)	48.1%	($\uparrow$ +3.5)	($\uparrow$ +4.9)	($\uparrow$ +1.0)	($\uparrow$ +5.3)
Llama3.1-8B	<i>Sparse</i>	66.6%	($\uparrow$ +2.5)	($\uparrow$ +3.0)	($\uparrow$ +2.2)	($\uparrow$ +3.7)	47.2%	($\uparrow$ +3.0)	($\uparrow$ +6.0)	($\uparrow$ +2.0)	($\uparrow$ +5.9)
	<i>Interval</i>	66.6%	($\uparrow$ +2.5)	($\uparrow$ +3.0)	($\uparrow$ +4.5)	($\uparrow$ +4.6)	47.2%	($\uparrow$ +3.0)	($\uparrow$ +6.0)	($\uparrow$ +4.0)	($\uparrow$ +6.6)
	<i>Dense</i>	66.6%	($\uparrow$ +2.5)	($\uparrow$ +3.0)	($\uparrow$ +4.8)	($\uparrow$ +3.7)	47.2%	($\uparrow$ +3.0)	($\uparrow$ +6.0)	($\uparrow$ +3.0)	($\uparrow$ +6.3)

Table 14: **Best-of-5 Reranking Performance & Baselines (%)**. Random is SFT pass@1. Deltas are percentage-point changes for each reranker. Blue arrows ($\uparrow$) indicate gains and purple arrows ($\downarrow$) indicate drops.

Backbone	Method	GSM8K					MMLU-Pro				
		Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.	Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.
Qwen2.5-4B	<i>Sparse</i>	76.6%	($\uparrow$ +0.1)	($\uparrow$ +8.3)	($\uparrow$ +9.1)	($\uparrow$ +9.1)	53.9%	($\uparrow$ +2.4)	($\uparrow$ +6.0)	($\uparrow$ +7.9)	($\uparrow$ +8.1)
	<i>Interval</i>	76.6%	($\uparrow$ +0.1)	($\uparrow$ +8.3)	($\uparrow$ +7.0)	($\uparrow$ +8.9)	53.9%	($\uparrow$ +2.4)	($\uparrow$ +6.0)	($\uparrow$ +1.9)	($\uparrow$ +5.8)
	<i>Dense</i>	76.6%	($\uparrow$ +0.1)	($\uparrow$ +8.3)	($\uparrow$ +7.7)	($\uparrow$ +9.1)	53.9%	($\uparrow$ +2.4)	($\uparrow$ +6.0)	($\uparrow$ +4.6)	($\uparrow$ +6.5)
Qwen2.5-7B	<i>Sparse</i>	70.1%	($\uparrow$ +1.3)	($\uparrow$ +13.0)	($\uparrow$ +10.5)	($\uparrow$ +13.9)	48.1%	($\uparrow$ +2.1)	($\uparrow$ +5.7)	($\uparrow$ +6.9)	($\uparrow$ +7.6)
	<i>Interval</i>	70.1%	($\uparrow$ +1.3)	($\uparrow$ +13.0)	($\uparrow$ +8.7)	($\uparrow$ +13.7)	48.1%	($\uparrow$ +2.1)	($\uparrow$ +5.7)	($\uparrow$ +3.2)	($\uparrow$ +6.2)
	<i>Dense</i>	70.1%	($\uparrow$ +1.3)	($\uparrow$ +13.0)	($\uparrow$ +11.0)	($\uparrow$ +14.4)	48.1%	($\uparrow$ +2.1)	($\uparrow$ +5.7)	($\uparrow$ +0.9)	($\uparrow$ +6.2)
Llama3.1-8B	<i>Sparse</i>	66.6%	($\uparrow$ +2.4)	($\uparrow$ +2.7)	($\uparrow$ +3.0)	($\uparrow$ +4.3)	47.2%	($\uparrow$ +2.5)	($\uparrow$ +7.0)	($\uparrow$ +1.4)	($\uparrow$ +6.6)
	<i>Interval</i>	66.6%	($\uparrow$ +2.4)	($\uparrow$ +2.7)	($\uparrow$ +5.3)	($\uparrow$ +4.8)	47.2%	($\uparrow$ +2.5)	($\uparrow$ +7.0)	($\uparrow$ +4.1)	($\uparrow$ +7.8)
	<i>Dense</i>	66.6%	($\uparrow$ +2.4)	($\uparrow$ +2.7)	($\uparrow$ +5.8)	($\uparrow$ +4.1)	47.2%	($\uparrow$ +2.5)	($\uparrow$ +7.0)	($\uparrow$ +3.7)	($\uparrow$ +7.4)

Table 15: **Best-of-8 Reranking Performance & Baselines (%)**. Random is SFT pass@1. Deltas are percentage-point changes for each reranker. Blue arrows ($\uparrow$) indicate gains and purple arrows ($\downarrow$) indicate drops.

Backbone	Method	GSM8K					MMLU-Pro				
		Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.	Random	Δ Logp.	Δ Maj.	Δ Rew.	Δ W.Maj.
Qwen2.5-4B	<i>Sparse</i>	76.6%	($\uparrow$ +1.4)	($\uparrow$ +8.7)	($\uparrow$ +10.3)	($\uparrow$ +10.6)	53.9%	($\uparrow$ +2.1)	($\uparrow$ +7.7)	($\uparrow$ +8.7)	($\uparrow$ +9.3)
	<i>Interval</i>	76.6%	($\uparrow$ +1.4)	($\uparrow$ +8.7)	($\uparrow$ +8.1)	($\uparrow$ +10.2)	53.9%	($\uparrow$ +2.1)	($\uparrow$ +7.7)	($\uparrow$ +2.2)	($\uparrow$ +7.5)
	<i>Dense</i>	76.6%	($\uparrow$ +1.4)	($\uparrow$ +8.7)	($\uparrow$ +9.2)	($\uparrow$ +10.6)	53.9%	($\uparrow$ +2.1)	($\uparrow$ +7.7)	($\uparrow$ +4.7)	($\uparrow$ +7.5)
Qwen2.5-7B	<i>Sparse</i>	70.1%	($\uparrow$ +0.8)	($\uparrow$ +15.6)	($\uparrow$ +11.3)	($\uparrow$ +16.4)	48.1%	($\uparrow$ +2.6)	($\uparrow$ +7.5)	($\uparrow$ +7.0)	($\uparrow$ +8.6)
	<i>Interval</i>	70.1%	($\uparrow$ +0.8)	($\uparrow$ +15.6)	($\uparrow$ +8.3)	($\uparrow$ +15.3)	48.1%	($\uparrow$ +2.6)	($\uparrow$ +7.5)	($\uparrow$ +4.1)	($\uparrow$ +8.1)
	<i>Dense</i>	70.1%	($\uparrow$ +0.8)	($\uparrow$ +15.6)	($\uparrow$ +11.1)	($\uparrow$ +15.7)	48.1%	($\uparrow$ +2.6)	($\uparrow$ +7.5)	($\uparrow$ +0.5)	($\uparrow$ +7.7)
Llama3.1-8B	<i>Sparse</i>	66.6%	($\uparrow$ +2.8)	($\uparrow$ +4.6)	($\uparrow$ +3.7)	($\uparrow$ +5.2)	47.2%	($\uparrow$ +2.7)	($\uparrow$ +7.6)	($\downarrow$ -0.2)	($\uparrow$ +8.3)
	<i>Interval</i>	66.6%	($\uparrow$ +2.8)	($\uparrow$ +4.6)	($\uparrow$ +5.9)	($\uparrow$ +6.8)	47.2%	($\uparrow$ +2.7)	($\uparrow$ +7.6)	($\uparrow$ +4.4)	($\uparrow$ +8.4)
	<i>Dense</i>	66.6%	($\uparrow$ +2.8)	($\uparrow$ +4.6)	($\uparrow$ +6.5)	($\uparrow$ +5.8)	47.2%	($\uparrow$ +2.7)	($\uparrow$ +7.6)	($\uparrow$ +3.2)	($\uparrow$ +7.9)

Table 16: **Best-of-16 Reranking Performance & Baselines (%)**. Random is SFT pass@1. Deltas are percentage-point changes for each reranker. Blue arrows ($\uparrow$) indicate gains and purple arrows ($\downarrow$) indicate drops.

D.5 Distribution Separation of Rewards

D.5.1 GSM8K

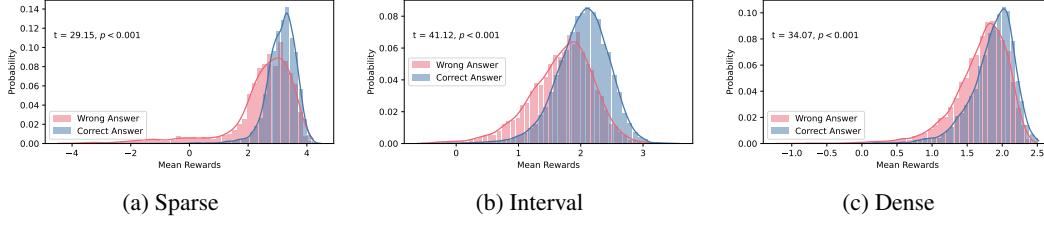


Figure 9: **Reward Distribution by Correctness (Qwen2.5-7B on GSM8K).** A t-test assesses the significance of the difference in mean rewards between correct and incorrect answers.

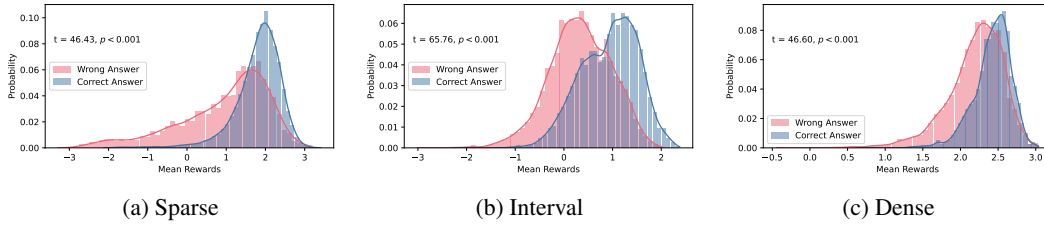


Figure 10: **Reward Distribution by Correctness (Llama3.1-8B on GSM8K).** A t-test assesses the significance of the difference in mean rewards between correct and incorrect answers.

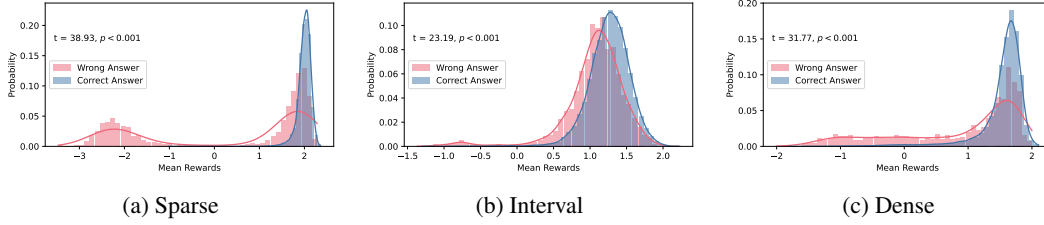


Figure 11: **Reward Distribution by Correctness (Qwen3-4B on GSM8K).** A t-test assesses the significance of the difference in mean rewards between correct and incorrect answers.

D.5.2 MMLU-PRO

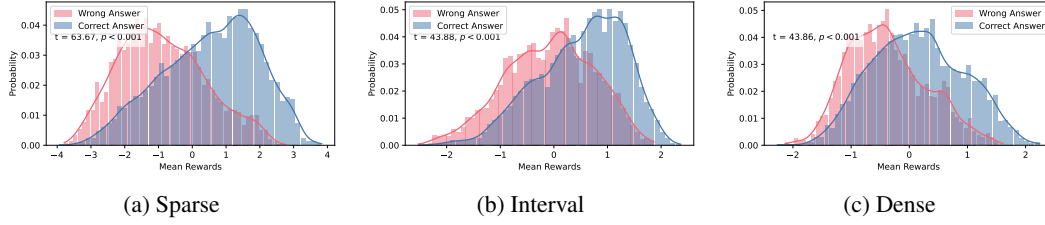


Figure 12: **Reward Distribution by Correctness (Qwen2.5-7B on MMLU-PRO)**. A t-test assesses the significance of the difference in mean rewards between correct and incorrect answers.

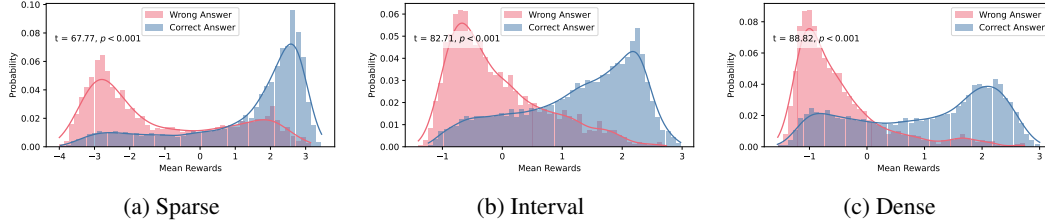


Figure 13: **Reward Distribution by Correctness (Llama3.1-8B on MMLU-PRO)**. A t-test assesses the significance of the difference in mean rewards between correct and incorrect answers.

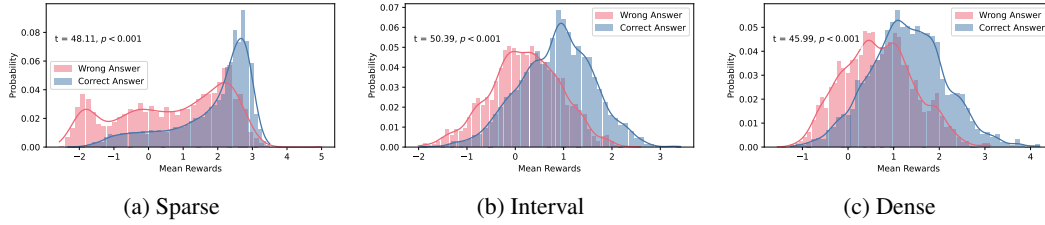


Figure 14: **Reward Distribution by Correctness (Qwen3-4B on MMLU-PRO)**. A t-test assesses the significance of the difference in mean rewards between correct and incorrect answers.

D.5.3 MEDREASON

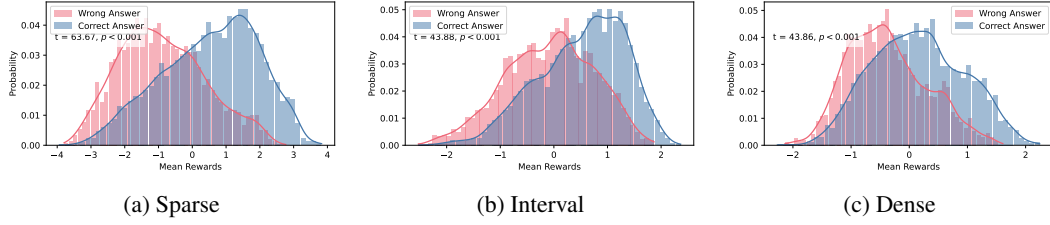


Figure 15: **Reward Distribution by Correctness (Qwen2.5-7B on MEDREASON)**. A t-test assesses the significance of the difference in mean rewards between correct and incorrect answers.

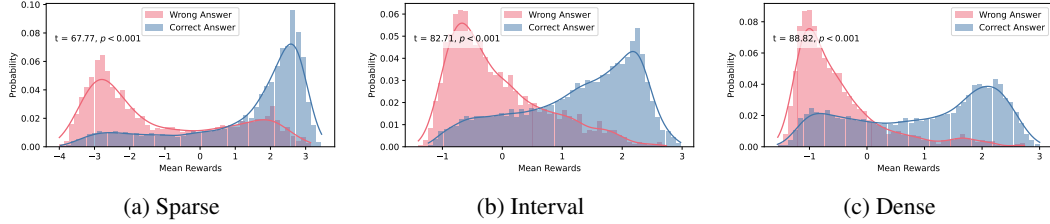


Figure 16: **Reward Distribution by Correctness (Llama3.1-8B on MEDREASON)**. A t-test assesses the significance of the difference in mean rewards between correct and incorrect answers.

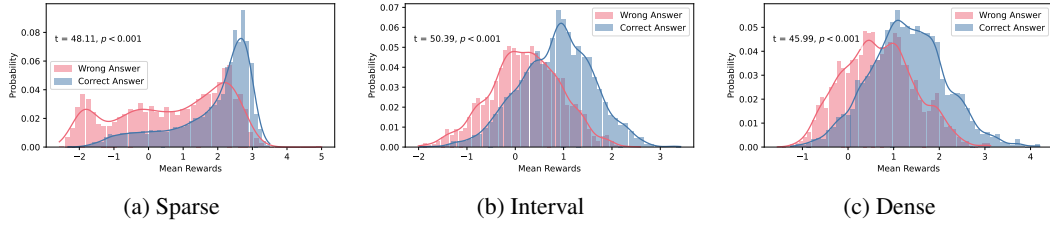


Figure 17: **Reward Distribution by Correctness (Qwen3-4B on MEDREASON)**. A t-test assesses the significance of the difference in mean rewards between correct and incorrect answers.

D.6 Error Localisation



Figure 18: **Error Localisation on GSM8K** Dense reward on correct and incorrect generations using the *dense* reasoning reward models. We clearly observe a divergence between the correct and incorrect paths.

D.7 Reasoning Traces

D.7.1 GSM8K

Question:

Kylar went to the store to buy glasses for his new apartment. One glass costs \$5, but every second glass costs only 60% of the price. Kylar wants to buy 16 glasses. How much does he need to pay for them?

Reasoning + Answer:

```
<think>
The deal is for every second glass, so for every pair, Kylar would need to pay: $5 + ($5 * 0.6) = $5 + $3 = $8
There are 16 / 2 = 8 pairs in 16 glasses
So Kylar needs to pay: 8 * 8 = $64 for them all.
</think>
<answer>
64
</answer>
```

(a) *Interval*, correct

Question:

Kylar went to the store to buy glasses for his new apartment. One glass costs \$5, but every second glass costs only 60% of the price. Kylar wants to buy 16 glasses. How much does he need to pay for them?

Reasoning + Answer:

```
<think>
One glass costs $5 and another one costs 60/100 * 5 = $3 cheaper.
So if every second glass costs only 60/100 * 5 = $3 cheaper, then the second glass costs 5 - 3 = $2
To buy 16 glasses Kylar needs to buy 2 * 8 = 16 glasses.
Kylar needs to pay 8 * 5 + 8 * 2 = 40 + 16 = $56 for all glasses.
</think>
<answer>
56
</answer>
```

(b) *Interval*, incorrect

Question:

Every day, Wendi feeds each of her chickens three cups of mixed chicken feed, containing seeds, mealworms and vegetables to help keep them healthy. She gives the chickens their feed in three separate meals. In the morning, she gives her flock of chickens 15 cups of feed. In the afternoon, she gives her chickens another 25 cups of feed. How many cups of feed does she need to give her chickens in the final meal of the day if the size of Wendi's flock is 20 chickens?

Reasoning + Answer:

```
<think>
In one feeding session Wendi gives each bird 3 cups of mixed feed, so she gives a total of 3 cups/bird * 20 birds = 60 cups of bird food.
In the morning she feeds her chickens 15 cups and in the afternoon she feeds them 25 cups, so she has 60 cups - 15 cups - 25 cups = 20 cups of bird food remaining for her last evening feeding.
</think>
<answer>
20
</answer>
```

(c) *Dense*, correct

Question:

Every day, Wendi feeds each of her chickens three cups of mixed chicken feed, containing seeds, mealworms and vegetables to help keep them healthy. She gives the chickens their feed in three separate meals. In the morning, she gives her flock of chickens 15 cups of feed. In the afternoon, she gives her chickens another 25 cups of feed. How many cups of feed does she need to give her chickens in the final meal of the day if the size of Wendi's flock is 20 chickens?

Reasoning + Answer:

```
<think>
The size of Wendi's flock of chicken is 20. She feeds each of her chicken in her flock 3 cups of chicken feed.
Wendi gives 15 cups of feed in the morning to her chickens and 25 cups in the afternoon.
She gives each chicken, 2 cups chicken feed twice in a day.
The total number of cups of feed given to the flock in a day is (2 * 20) = 40.
Wendi needs 40 cups of chicken feed less the 40 cups she feeds her flock in a day in the final meal of the day which is 75 cups.
</think>
<answer>
75
</answer>
```

(d) *Dense*, incorrect

Figure 19: Correct and Incorrect Reasoning Reward for Qwen2.5-7B on GSM8K. Dense reward on correct and incorrect generations using the *interval* and *dense* reasoning reward model.

Question:

Every day, Wendi feeds each of her chickens three cups of mixed chicken feed, containing seeds, mealworms and vegetables to help keep them healthy. She gives the chickens their feed in three separate meals. In the morning, she gives her flock of chickens 15 cups of feed. In the afternoon, she gives her chickens another 25 cups of feed. How many cups of feed does she need to give her chickens in the final meal of the day if the size of Wendi's flock is 20 chickens?

Reasoning + Answer:

```
<think>
Thus far today, Wendi has given her chickens 15 + 25 = 40 cups of feed.
She needs to give her 20 chickens 3 cups of feed each, or 20 * 3 = 60 cups of feed total for the day.
Therefore, Wendi needs to give her chickens 60 - 40 = 20 cups of feed in the final meal of the day.
</think>
<answer>
20
</answer>
```

(a) *Interval*, correct

Question:

Every day, Wendi feeds each of her chickens three cups of mixed chicken feed, containing seeds, mealworms and vegetables to help keep them healthy. She gives the chickens their feed in three separate meals. In the morning, she gives her flock of chickens 15 cups of feed. In the afternoon, she gives her chickens another 25 cups of feed. How many cups of feed does she need to give her chickens in the final meal of the day if the size of Wendi's flock is 20 chickens?

Reasoning + Answer:

```
<think>
Thus far, Wendi has given her chickens 15 + 25 = 40 cups of feed.
Since she feeds each chicken three cups of feed per meal, she has already fed each chicken 40 / 3 = 13 and 1/3 cups of feed.
Therefore, to complete the three meals per day, she must feed each chicken an additional 3 - 13 and 1/3 = -10 and 1/3 cups of feed.
Since a chicken cannot be fed a negative amount of feed, for the final meal of the day, Wendi will need to feed each chicken an additional 10 and 1/3 cups of feed.
For the entire flock of 20 chickens, she will need to feed them 20 * 10 and 1/3 = 200 and 2/3 cups of feed.
Thus, for the final meal of the day, Wendi will need to give her chickens 200 and 2/3 cups of feed.
</think>
<answer>
200
</answer>
```

(b) *Interval*, incorrect

Question:

Every day, Wendi feeds each of her chickens three cups of mixed chicken feed, containing seeds, mealworms and vegetables to help keep them healthy. She gives the chickens their feed in three separate meals. In the morning, she gives her flock of chickens 15 cups of feed. In the afternoon, she gives her chickens another 25 cups of feed. How many cups of feed does she need to give her chickens in the final meal of the day if the size of Wendi's flock is 20 chickens?

Reasoning + Answer:

```
<think>
Thus far today, Wendi has given her chickens 15 + 25 = 40 cups of feed.
She needs to give her 20 chickens 3 cups of feed each, or 20 * 3 = 60 cups of feed total for the day.
Therefore, Wendi needs to give her chickens 60 - 40 = 20 cups of feed in the final meal of the day.
</think>
<answer>
20
</answer>
```

(c) *Dense*, correct

Question:

Every day, Wendi feeds each of her chickens three cups of mixed chicken feed, containing seeds, mealworms and vegetables to help keep them healthy. She gives the chickens their feed in three separate meals. In the morning, she gives her flock of chickens 15 cups of feed. In the afternoon, she gives her chickens another 25 cups of feed. How many cups of feed does she need to give her chickens in the final meal of the day if the size of Wendi's flock is 20 chickens?

Reasoning + Answer:

```
<think>
Thus far in the day, Wendi has fed her chickens 15 cups + 25 cups = 40 cups of chicken feed.
Therefore, for the final meal of the day, she needs to feed her 20 chickens 40 cups / 3 meals = 13.33 cups per chicken.
Thus, Wendi needs to give her chickens 13.33 * 20 = 266.60 cups of feed.
For the sake of the chickens, she will need to round up to the nearest whole cup, meaning she needs to give her chickens 267 cups of chicken feed in the final meal of the day.
</think>
<answer>
267
</answer>
```

(d) *Dense*, incorrect

Figure 20: Correct and Incorrect Reasoning Reward for Llama3.1-8B on GSM8K. Dense reward on correct and incorrect generations using the *interval*, and *dense* reasoning reward model.

D.7.2 MMLU-Pro

Question:

Which of the following factors is associated with a decreased risk of Alzheimer's?

Answer Choices:

- A. Being African or Hispanic American
- B. A lower level of education
- C. Living in a city
- D. Being married
- E. Having a family history of Alzheimer's
- F. High intake of saturated fats
- G. Older age
- H. Eating fish
- I. Being a smoker
- J. Drinking alcohol regularly

Reasoning + Answer:

<think>

I can do a step-by-step chain-of-thought, but here is a concise explanation of why the correct answer is as stated.

Individual risk factors for Alzheimer's disease are mostly immutable (age, sex, and family history). Demographic and lifestyle factors that generally confer higher risk include older age, lower education (which often reflects broader cognitive reserve that is protective), having African or Hispanic ancestry (higher risk), living in urban environments (often higher stress), smoking, and heavy alcohol use. Alcohol has complicated associations: moderate drinking can be neuroprotective in some observational data, but chronic or binge patterns increase cognitive risk. Regularly eating fish has been consistently associated with a lower risk of cognitive decline and Alzheimer's disease (possibly through omega-3 fatty acids, anti-inflammatory effects, and mental stimulation from engaging with the activity); this stands in contrast to the other options, which typically raise risk.

Correct answer: H. Eating fish

</think>

<answer>

H. Eating fish

</answer>

(a) *Interval*, correct

Question:

Which of the following factors is associated with a decreased risk of Alzheimer's?

Answer Choices:

- A. Being African or Hispanic American
- B. A lower level of education
- C. Living in a city
- D. Being married
- E. Having a family history of Alzheimer's
- F. High intake of saturated fats
- G. Older age
- H. Eating fish
- I. Being a smoker
- J. Drinking alcohol regularly

Reasoning + Answer:

<think>

Alzheimer's disease pathology is profoundly influenced by vascular factors, with vascular dysfunction (hypertension, diabetes, hyperlipidemia) raising amyloid β deposition and tangles. Education is a robust correlate of lower Alzheimer's risk because higher cognitive reserve and greater years of education are associated with a survival advantage. Survivors experience early cognitive symptoms and functional decline, while non-survivors show faster progression. This relationship with education is distinct from mere age (option G), because younger individuals with higher education have longer remaining cognitive reserve at the same age and are thus less likely to develop overt clinical disease.

By contrast:

- Genetics (e.g., APOE ϵ 4) and family history (E) increase risk.
- Advanced age (G) is a risk factor.
- African or Hispanic ancestry is not, on average, protective in U.S. populations; in fact, higher lifetime risk exists (A).
- Smoking (I) and alcohol (J) are risk factors.
- High saturated fat intake (F) and lower education (B) are risk factors, consistent with vascular risk models and neuroprotective cognitive reserve.
- Living in a city is not associated with a consistent risk change in Alzheimer's incidence.
- Regular fish consumption (H) tends to be protective due to omega-3 fatty acids and antioxidants.

Correct answer: D. Being married

Explanation: Marital status moderates mortality risk and care support, influencing survival and clinical detection; older adults with a history of marriage tend to survive their baseline neurocognitive change longer. Therefore, from a cohort surviving long enough to experience clinical disease, those who were previously married are more likely to have clinical Alzheimer's than never married peers.

However, the reasoning actually points to education as the factor associated with a decreased risk. The key discrepancy is in the phrasing of the correct option versus the actual factor highlighted in the explanation. To conform strictly to the provided options, the intended selection is D. Being married.

</think>

<answer>

D. Being married

</answer>

(b) *Interval*, incorrect

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
- B. 2.8 amps, 3.2 ohms
- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

To dissipate 40 watts with a 12 V battery, use $P = V^2 / R$: $40 \text{ W} = (12 \text{ V})^2 / R \implies R = 144 / 40 = 3.6 \text{ } \Omega$.

Current drawn from a 12 V source with $R = 3.6 \text{ } \Omega$ is $I = V / R = 12 / 3.6 = 3.33 \text{ A}$.

Therefore, the correct choice is E. 3.33 amps, 3.6 ohms.

</think>

<answer>

E. 3.33 amps, 3.6 ohms

</answer>

(c) *Dense*, correct

Question:

An automobile battery produces a potential difference (or "voltage") of 12 volts between its terminals. (It really consists of six 2 volt batteries following one after the other.) A headlightbulb is to be connected directly across the terminals of the battery and dissipate 40 watts of joule heat. What current will it draw and what must its resistance be?

Answer Choices:

- A. 3.0 amps, 4 ohms
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- C. 3.75 amps, 3.2 ohms
- D. 4 amps, 3 ohms
- E. 3.33 amps, 3.6 ohms
- F. 5 amps, 2.4 ohms
- G. 4.5 amps, 2.67 ohms
- H. 2.2 amps, 5.45 ohms
- I. 2.5 amps, 4.8 ohms
- J. 3.5 amps, 4 ohms

Reasoning + Answer:

<think>

- The bulb's rated power and voltage determine its resistance: $P = V^2 / R \implies R = V^2 / P = (12 \text{ V})^2 / 40 \text{ W} = 36 / 40 = 0.9 \text{ } \Omega$.
- With a 12 V battery and $R = 0.9 \text{ } \Omega$, the current drawn is $I = V / R = 12 \text{ V} / 0.9 \text{ } \Omega \approx 13.33 \text{ A}$.
- This exceeds typical automotive practical limits (e.g., short circuits), so use the given choices. The stated bulb wattage of 40 W is highly implausible for a 12 V system, as a 12 V / 40 W bulb would require $R \approx 0.9 \text{ } \Omega$, which gives $I \approx 13.33 \text{ A}$ too high.

Therefore, the correct answer, based on the standard calculation and implied unlikelihood of the stated power, is: C. 3.75 amps, 3.2 ohms.

Correct Answer: C.

</think>

<answer>

C. 3.75 amps, 3.2 ohms

</answer>

(d) *Dense*, incorrect

Figure 21: Correct and Incorrect Reasoning Reward for Qwen2.5-7B on MMLU-PRO. Dense reward on correct and incorrect generations using the *interval* and *dense* reasoning reward model.

Question:
Which of the following factors is associated with a decreased risk of Alzheimer's?

Answer Choices:
A. Being African or Hispanic American
B. A lower level of education
C. Living in a city
D. Being married
E. Having a family history of Alzheimer's
F. High intake of saturated fats
G. Older age
H. Eating fish
I. Being a smoker
J. Drinking alcohol regularly

Reasoning + Answer:
<think>
Alzheimer's disease shows clear sociodemographic and lifestyle risk-modifying patterns.
- Education: Higher educational attainment is consistently associated with reduced risk of Alzheimer's pathology and clinically evident dementia, including APOE ε4 carriers. Cognitive reserve enhances resilience to neurodegenerative processes, especially late in life when pathologic burden increases. Thus, option A (a lower level of education) is not correct.
- Ethnicity: Alzheimer's incidence is lower in African Americans than in non-Hispanic whites, though they have a higher age-adjusted prevalence due to increased longevity. Alzheimer's tends to be underdiagnosed in African American communities. Hispanics/Latinos also appear to have lower risk before late life, though cumulative incidence rises with age. Therefore, being African or Hispanic American is not associated with decreased risk; it is closer to neutral (or sometimes lower) risk, and this option is ambiguous as presented.
- Marital status: Widowhood and marital loss are linked to higher risk of dementia, likely through increased psychosocial stress, isolation, lack of cognitive stimulation, and reduced healthcare access. Being married is generally associated with lower risk.
- City living vs rural: Some evidence suggests rural residence is associated with lower risk, whereas urban environments are linked to higher dementia incidence (e.g., through cumulative stress, less green space, urban heat island effects, poor air quality, and neighborhood disadvantage). Thus, living in a city is not protective.
- Family history: A positive family history markedly increases risk, especially for autosomal dominant forms (e.g., early-onset Alzheimer's (FAD) and for late-onset disease overall. Having a family history is therefore not associated with decreased risk.
- Saturated fat intake: Data do not show a protective effect of higher saturated fat consumption; healthier dietary patterns (e.g., the MIND diet emphasizing fruits, vegetables, nuts, whole grains, and healthy fats, with less saturated fat) correlate with lower risk, but high saturated fat intake is not itself protective.
- Fish intake: Omega-3 fatty acids in fish (particularly EPA and DHA) show mixed but generally supportive evidence for cognitive protection, with lower risk of cognitive decline and dementia observed in some longitudinal studies. Thus, eating fish is associated with decreased risk.
- Older age: Age is the strongest risk factor for Alzheimer's; the risk rises substantially after age 65, with most cases occurring in individuals over 65. Therefore, older age is not associated with a decreased risk.
- Smoking: Tobacco use is strongly harmful to cognition and increases dementia risk; it is not protective.
- Alcohol: Light, regular alcohol consumption may be neuroprotective in some contexts (potentially via anti-inflammatory effects and improved cardiovascular risk), but this is not uniformly protective, can be harmful with excess, and is not a primary or reliable protective factor for most people.
Evaluating the options, the factor most consistently associated with decreased risk of Alzheimer's is a higher intake of fish, through its omega-3 fatty acid content.
Correct answer: H. Eating fish
</think>
<answer>
H. Eating fish
</answer>

(a) *interval*, correct

Question:
At birth, the least developed part of the brain is the

Answer Choices:
A. visual system
B. thalamus
C. frontal lobe
D. brain stem
E. parietal lobe
F. cerebellum
G. cortex
H. basal ganglia
I. hypothalamus
J. limbic system

Reasoning + Answer:
<think>
Answer: G. cortex
Explanation (concise):
- The cerebral cortex, including its six layers, is largely immature at birth. Cortical synapses and connectivity (especially long-range and horizontal associations) develop postnatally through experience-dependent pruning and myelination.
- In contrast, subcortical structures such as the thalamus, hypothalamus, basal ganglia, brain stem, and cerebellum have functional rudiments present at birth (e.g., visual reflex arcs, tone regulation, early motor coordination), even though they continue to mature.
- Sensory systems (visual, auditory) show early functional beginnings; while the visual system is relatively underdeveloped at birth, it is not the least developed overall in terms of gross neural structure and functional integration.
- The limbic system has early organizing influences, especially the hippocampus for memory traces, but its systems-level functional maturation advances more than the cortex in the first weeks of life.
Therefore, the least developed part of the brain at birth is the cortex.
Correct answer: G. cortex
</think>
<answer>
G. cortex
</answer>

(c) *Dense*, correct

Question:
Which of the following factors is associated with a decreased risk of Alzheimer's?

Answer Choices:
A. Being African or Hispanic American
B. A lower level of education
C. Living in a city
D. Being married
E. Having a family history of Alzheimer's
F. High intake of saturated fats
G. Older age
H. Eating fish
I. Being a smoker
J. Drinking alcohol regularly

Reasoning + Answer:
<think>
Answer: E. Having a family history of Alzheimer's
Concise explanation:
Family history of Alzheimer disease increases an individual's risk due to shared genetic predispositions (for example, APOE ε4). In contrast, higher educational attainment is consistently associated with lower incidence and slower progression of Alzheimer pathology, reflecting increased cognitive reserve. The other listed factors either have weak, inconsistent, or no protective effect and are not established risk reductions in the way higher education is. Therefore, the factor most clearly associated with a decreased risk is having a family history of Alzheimer's, which is a positive risk factor rather than a protective one. I apologize for the initial word choice in the prompt.
Correct Answer: E. Having a family history of Alzheimer's
</think>
<answer>
E. Having a family history of Alzheimer's
</answer>

(b) *interval*, incorrect

Question:
At birth, the least developed part of the brain is the

Answer Choices:
A. visual system
B. thalamus
C. frontal lobe
D. brain stem
E. parietal lobe
F. cerebellum
G. cortex
H. basal ganglia
I. hypothalamus
J. limbic system

Reasoning + Answer:
<think>
Developmental neuroanatomy shows that the brain proceeds from caudal (tail) to rostral (head) and from general/ arousal systems to specialized associations and cortices. The brain stem (mid brain, pons, medulla) contains critical life-support structures and autonomic centers, respiratory networks, vestibular nuclei, reticular formation, and cranial nerve nuclei that must function early in development. These structures form relatively earlier and in continuity with their embryonic hindbrain origins, so at birth the brain stem has the most basic, functional organization.
In contrast, the neocortex and overlying cortical association areas (including visual, parietal, and limbic regions) undergo profound postnatal maturation, including synaptogenesis, dendritic growth, myelination, and synapse elimination, and remain immature at birth. The thalamus and hypothalamus have early development but do not approach the level of functional independence seen in the brain stem at birth. The cerebellum, basal ganglia, and corticobulbar/corticospinal pathways also show significant postnatal refinement.
Because the least developed region at birth is the neocortex and by extension its specialized components such as the visual system, this is the correct choice.
Correct answer: A. visual system
</think>
<answer>
A. visual system
</answer>

(d) *Dense*, incorrect

Figure 22: Correct and Incorrect Reasoning Reward for Llama3.1-8B on MMLU-PRO. Dense reward on correct and incorrect generations using the *interval* and *dense* reasoning reward model.

D.7.3 MedReason

Question:

Shock causes:

Answer Choices:

A. Stagnant hypoxia

B. Anemic hypoxia

C. Hypoxic hypoxia

D. Histotoxic hypoxia

Reasoning + Answer:

mobx

Shock leads to inadequate blood flow to tissues, resulting in insufficient oxygen delivery to cells. This is primarily due to reduced perfusion, not a decrease in oxygen content of the blood or impaired cellular utilization.

Let AGIs evaluate each option:

A. Stagnant hypoxia AGI This occurs when blood flow to tissues is reduced (e.g., due to poor perfusion), leading to oxygen deprivation despite adequate oxygen content in the blood. This is exactly what happens in shock, where circulation is impaired and blood flow is stagnant. This is the most appropriate answer.

B. Anemic hypoxia AGI This results from a decrease in hemoglobin or red blood cells (e.g., hemorrhage, hemolytic anemia), leading to reduced oxygen-carrying capacity. While hemorrhagic shock may involve anemia, the primary mechanism of hypoxia in shock is not anemia per se, but rather poor perfusion.

C. Hypoxic hypoxia AGI This occurs when oxygen content in the blood is low due to low partial pressure of oxygen (e.g., in pulmonary disorders). This is not the primary mechanism in shock.

D. Histotoxic hypoxia AGI This occurs when cells cannot utilize oxygen (e.g., cyanide poisoning). It is not related to shock.

In shock, the main problem is that tissues receive insufficient blood flow, leading to stagnant hypoxia. The blood may have normal oxygen content, but it's not reaching the tissues due to poor circulation.

</think>

A. Stagnant hypoxia

</answer>

(a) Interval, correct

Question:

Shock causes:

Answer Choices:

A. Stagnant hypoxia

B. Anemic hypoxia

C. Hypoxic hypoxia

D. Histotoxic hypoxia

Reasoning + Answer:

mobx

Shock leads to inadequate tissue perfusion, resulting in reduced oxygen delivery to tissues. The primary issue is that oxygen cannot reach tissues adequately due to poor blood flow, even if oxygen content in the blood is normal. This condition is classified as hypoxic hypoxia, where the tissue is deprived of oxygen because of insufficient oxygen delivery due to impaired perfusion.

Let AGIs evaluate the options:

A. Stagnant hypoxia AGI This occurs when blood flow is reduced but oxygen content is normal, and the tissue is hypoxic due to poor blood flow. It is a form of hypoxic hypoxia and can be seen in shock, but the term "stagnant hypoxia" is more specific to reduced blood flow without oxygen content change. However, in shock, the main issue is reduced delivery, which is more precisely described as hypoxic hypoxia.

B. Anemic hypoxia AGI This occurs due to reduced hemoglobin (e.g., hemorrhage), leading to decreased oxygen-carrying capacity. While hemorrhagic shock can cause anemia, the core pathophysiology of shock is not primarily anemia per se.

C. Hypoxic hypoxia AGI This is defined by reduced oxygen tension in the tissues due to inadequate oxygen delivery from the blood, typically due to poor perfusion. This is the hallmark of shock. The tissues are hypoxic because blood flow is insufficient, even if oxygen content is normal.

D. Histotoxic hypoxia AGI This occurs when cells cannot utilize oxygen (e.g., cyanide poisoning), not due to delivery issues.

In shock, the primary mechanism is reduced perfusion leading to inadequate oxygen delivery to tissues, which is the definition of hypoxic hypoxia.

</think>

C. Hypoxic hypoxia

</answer>

(b) Interval, incorrect

Figure 23: Correct and Incorrect Reasoning Reward for Qwen3-4B on MEDREASON. Dense reward on correct and incorrect generations using the *dense* reasoning reward model.

Question:

Shock causes:

Answer Choices:

A. Stagnant hypoxia

B. Anemic hypoxia

C. Hypoxic hypoxia

D. Histotoxic hypoxia

Reasoning + Answer:

<think>

To determine the correct answer, let's analyze the options given. Shock can be caused by various factors, including inadequate blood flow, leading to insufficient oxygen delivery to tissues. Stagnant hypoxia (A) occurs when there is a lack of blood flow due to cardiac dysfunction, resulting in inadequate oxygen delivery to tissues. This is a common cause of shock, particularly in conditions such as cardiogenic shock.

Anemic hypoxia (B) is caused by a decrease in the amount of hemoglobin or red blood cells, which reduces the capacity of the blood to carry oxygen. While this can lead to tissue hypoxia, it is not the primary cause of shock.

Hypoxic hypoxia (C) occurs when there is a decrease in the partial pressure of oxygen in the air or in the blood. This can be caused by respiratory or altitude-related issues but is not directly related to the causes of shock.

Histotoxic hypoxia (D) is caused by the presence of substances that interfere with the body's ability to use oxygen, such as certain toxins. While this can lead to tissue hypoxia, it is not a direct cause of shock.

Considering the options, stagnant hypoxia is a direct cause of shock due to its impact on blood flow and oxygen delivery to tissues.

</think>

<answer>

A. Stagnant hypoxia

</answer>

(a) interval, correct

Question:

Shock causes:

Answer Choices:

A. Stagnant hypoxia

B. Anemic hypoxia

C. Hypoxic hypoxia

D. Histotoxic hypoxia

Reasoning + Answer:

<think>

To answer this question, we need to identify the cause of shock. Shock is a systemic condition characterized by inadequate perfusion of vital organs, leading to organ dysfunction. It can result from various causes, including inadequate oxygen delivery to the tissues (hypoxia).

A. Stagnant hypoxia refers to a situation where there is impaired blood flow, leading to inadequate oxygen delivery to tissues. This can be caused by cardiac arrest, severe cardiac dysfunction, or significant blood loss. However, this is not the primary cause of shock.

B. Anemic hypoxia occurs when the blood lacks sufficient red blood cells or hemoglobin, impairing oxygen delivery to tissues. This can be caused by anemia, severe bleeding, or carbon monoxide poisoning. While anemia can contribute to shock, it is not the most direct cause.

C. Hypoxic hypoxia refers to inadequate oxygen levels in the air breathed in (hypoxia) or a failure of the lungs to exchange oxygen properly. This can be caused by high altitudes, respiratory failure, or carbon monoxide poisoning. Hypoxic hypoxia is a direct cause of shock, as it impairs the body's ability to deliver oxygen to tissues.

D. Histotoxic hypoxia occurs when certain toxins or substances interfere with the body's ability to use oxygen. This can be caused by carbon monoxide poisoning, cyanide poisoning, or other toxins. While histotoxic hypoxia can contribute to shock, it is not the primary cause.

Considering the options, the most direct cause of shock is related to inadequate oxygen delivery to tissues, which is best described by option C. Hypoxic hypoxia is a direct cause of shock, as it impairs the body's ability to deliver oxygen to tissues.

</think>

<answer>

C. Hypoxic hypoxia

</answer>

(b) interval, incorrect

Figure 24: Correct and Incorrect Reasoning Reward for Llama3.1-8B on MEDREASON. Dense reward on correct and incorrect generations using the *interval* reasoning reward model.